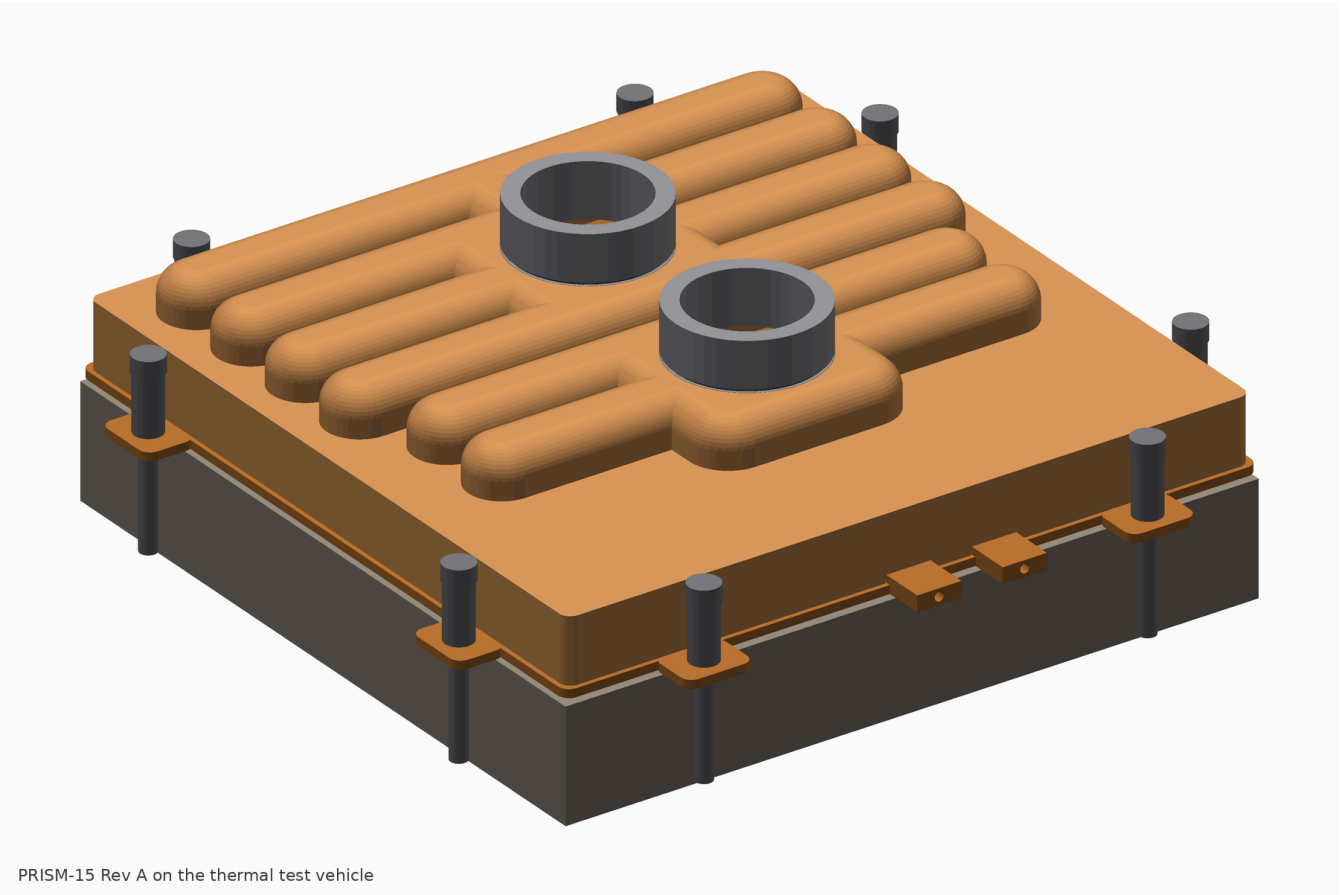


PRISM-15 Rev A on the thermal test vehicle

DESIGN REPORT

PRISM-15 Rev A

Cold-plate design report

16 October 2026
Benchmark: CoolIT Systems 15 kW cold plate (tech brief, June 2026)
Status: design definition complete for every layer; CAD regenerated and checked; no hardware yet
Source: docs/cooling in the Rez repository

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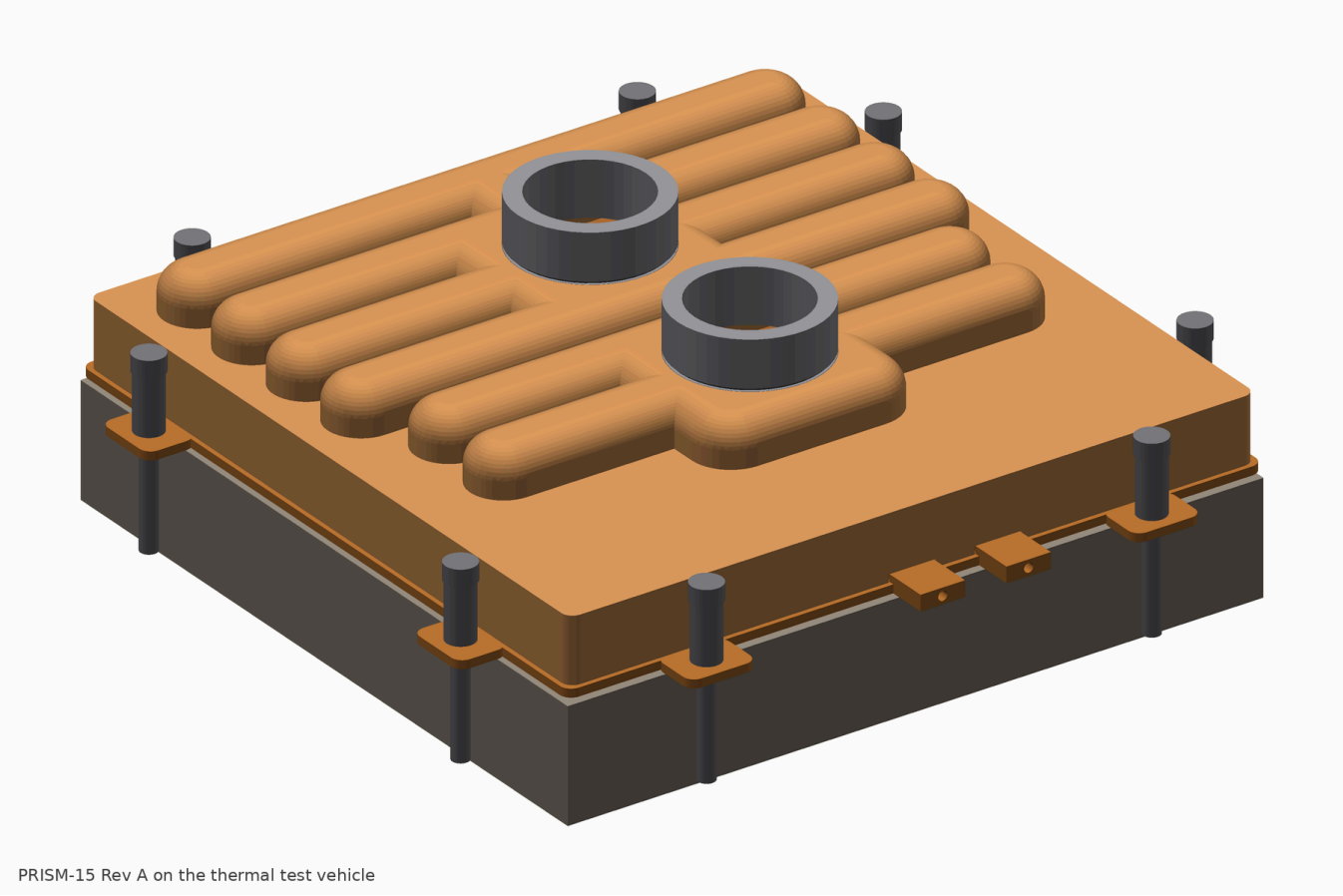
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0. Executive summary

What this is. PRISM-15 Rev A is a single-phase, all-copper, direct-liquid-cooled cold plate designed to beat CoolIT Systems' 15 kW cold plate (tech brief, June 2026) at CoolIT's own operating point and in CoolIT's own units. This report is the design package: the architecture of record carried through to a dimensioned design, a parametric CAD model with STEP, STL and 2-D artwork files, the manufacturing route, the predicted datasheet, and the programme that turns the model into measured hardware.

How it works, in three sentences. A split-flow plate sends every parcel of coolant 54 mm along the fins, so the far edge runs on coolant that has already warmed by 12.7 K, and the long path forces coarse 0.2 mm fins to hold pressure drop down. PRISM feeds a fine skived fin field (0.15 mm fins at 0.15 mm gaps) through fifteen metered lines, so every parcel travels 7.7 mm, every fin segment starts at inlet temperature, and the fin geometry is chosen for heat transfer and fouling rather than for pressure drop. The fin tips are brazed to the manifold grate, so a 0.85 mm base carries the coolant pressure without bowing and holds a phase-change pad or liquid metal at its rated resistance.



PRISM-15 Rev A on the thermal test vehicle

Scorecard at CoolIT's test point (15 kW, 18 LPM PG25, 45 °C inlet, 117 cm² heated area; PRISM values are model predictions, ±25 % per term until tested).

Metric	CoolIT 15 kW	PRISM-15 Rev A	Verdict
Average case-to-inlet rise, same paste class	34.4 K (measured)	31.0 K	win, P ≈ 0.8
Maximum case-to-inlet rise, same paste class	44.2 K (measured)	38.7 K	win, P ≈ 0.9

Metric	CoolIT 15 kW	PRISM-15 Rev A	Verdict
Average / maximum on a phase-change pad	≈ 32.5 / 40.3 K (estimate)	29.0 / 34.8 K	win
Average / maximum with liquid metal (Tier 2)	–	23.9 / 28.8 K	stretch
Plate pressure drop at 18 LPM, 20 mm ports	undisclosed; ≈ 22 kPa reconstructed	7.6 kPa	win
Load at 85 °C case, 1.2 LPM/kW, paste / pad	12.9 / 14.8 kW	15.7 / 18.5 kW	win
Inlet temperature allowed at 15 kW, paste / pad	40.8 / 44.7 °C	46.3 / 50.2 °C	win
Height at the headers / mass	≈ 22–28 mm / 1.2–1.6 kg (estimate)	20.8 mm / 1.3 kg	parity to win
Factory cost per plate at 100k/yr	≈ US$126	US$165	loss (cost per kW at parity)

What it does not win. Price per plate (+31 %), field record, fouling life at equal filtration (it ships with a 33 µm tray filter), the average temperature at very low flow or if CoolIT's heaters were only 100 cm², and CoolIT's small 4 kW single-die point with the drop-in core. These are stated in full in section 10.

Design status. The design definition is complete for every layer: core, grate, plenum and separator, cap and bosses, mounting, manufacturing route, predicted datasheet and innovation register. The CAD assembly of fourteen parts is regenerated from the lane values and checked for interference (section 9). The cap lane closed the grate-outline conflict at 121.0 mm; the datum-pin diameter, the ear-hole size, the mass recovery and the guard variant are the items left for the Rev A freeze at gate G1. No hardware exists yet: the first braze coupons, a water-table flow test and purchased incumbent plates are the first three months of the programme.

Decisions requested (detail in section 13): approve the scoped public claim rather than "beats on every metric"; fund in tranches, US$2.3 M to gate G1 at month 3; adopt the vacuum-braze Rev A build with the silver-free Cu-Sn joint decided at G1; make the 33 µm tray filter a supplied part; hold US sales until the freedom-to-operate opinion clears.

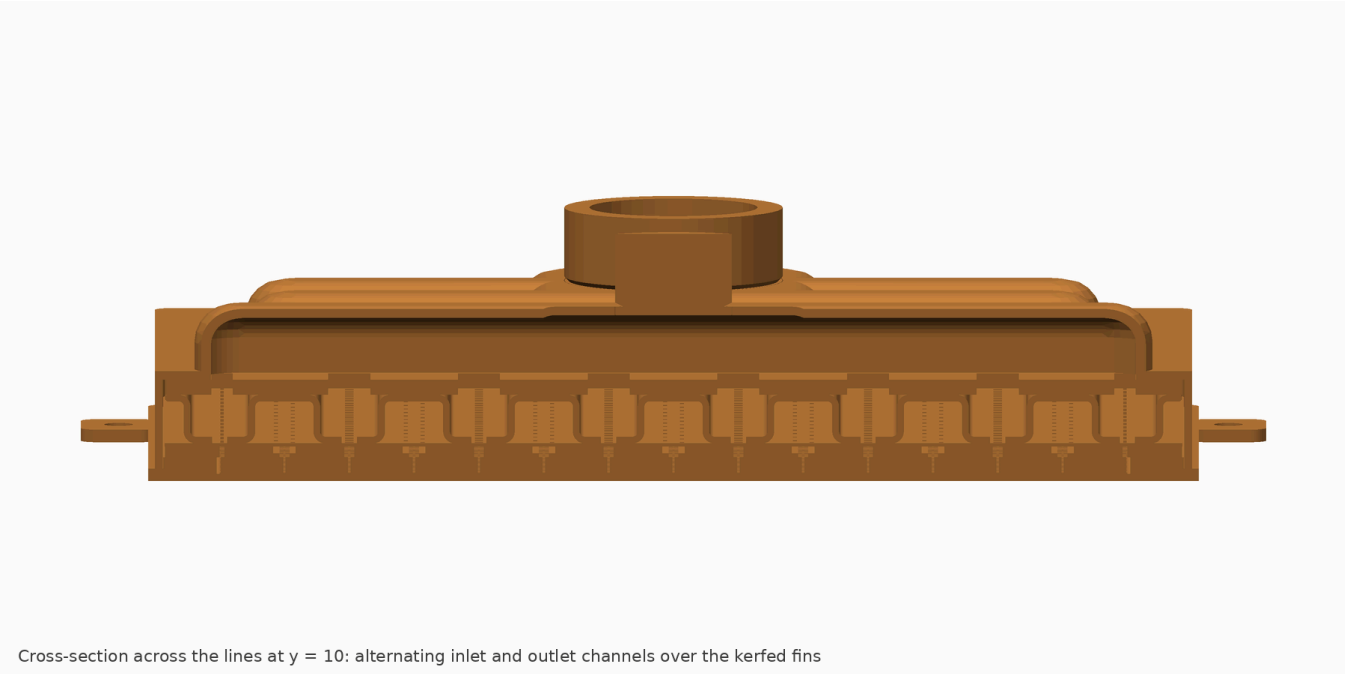
1. The benchmark and the rules of the comparison

CoolIT's brief reports one operating point: 15 kW from four ceramic heaters, 18 LPM of water-glycol at 45 °C inlet, three plates each read by eight thermocouples at the heater surface, standard thermal paste. The averages are 34.4 K case-to-inlet (plate means 32.5, 35.2 and 35.6 K) and the hottest reading is 44.2 K. The brief does not give the pressure drop, the heated area, the paste identity, the height, the mass or the thermal resistance against flow.

Our reconstruction, built in the research phase and documented in `docs/cooling/01-benchmark-and-requirements.md`, puts the heated area at about 120 cm² (photogrammetry of the brief's own photographs, 100 to 145 cm² band), which makes the plate a 0.27 cm²·K/W device at 128 W/cm². A one-dimensional model calibrated to that footprint reproduces the measured 34.4 K and decomposes it as coolant rise 6.3 K, paste 9.0 K, 2 mm base 6.6 K and convection 12.6 K. The plate pressure drop is bounded at about 22 kPa (17 to 42 kPa band) from the fin geometry a skived split-flow plate must have to reach that convection term.

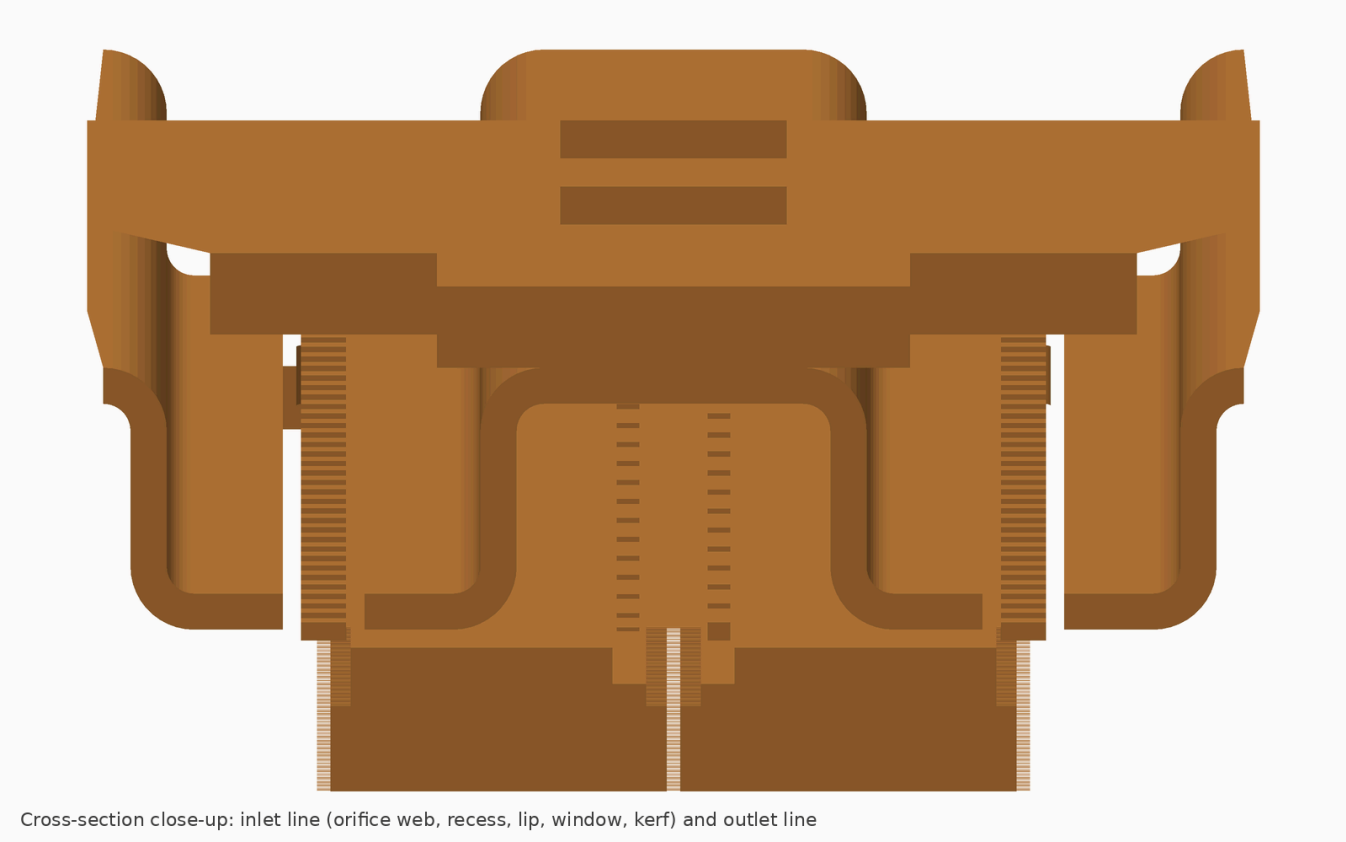
The arbiter for the public claim is the OCP Single-Phase Direct-to-Chip Cold Plate Test Specification v1.0.0 (effective September 2026), whose primary author also wrote CoolIT's brief. Every PRISM number in this report is therefore stated the way that specification measures: case-to-inlet resistance with the thermal interface included, at a stated flow, with the pressure drop at a stated boundary. The comparison rules we impose on ourselves: paste and pad results published separately with the measured bond line; two test vehicles (117 cm² parity and 100 cm² hard mode); the eight-thermocouple layout replicated; pressure drop at three boundaries (port-to-port with bore stated, with barbs, with a named quick-disconnect pair); five plates by three re-mounts with an independent ISO/IEC 17025 repeat.

2. The mechanism



Cross-section across the lines at y = 10: alternating inlet and outlet channels over the kerfed fins

The cross-section above, taken from the CAD assembly across the fifteen lines, shows the whole mechanism in one picture. From the bottom: the 0.85 mm base and the 2.0 mm fin field, cut by 0.3 mm kerfs at every line; the 1.3 mm grate land plate with a 1.2 mm window over each kerf and, at inlet lines, a 2.7 mm recess; the 0.4 mm meter web with its graded orifices; the folded plenum whose channels alternate inlet and outlet at 7.7 mm pitch; the separator; and the drawn cap with its arches.

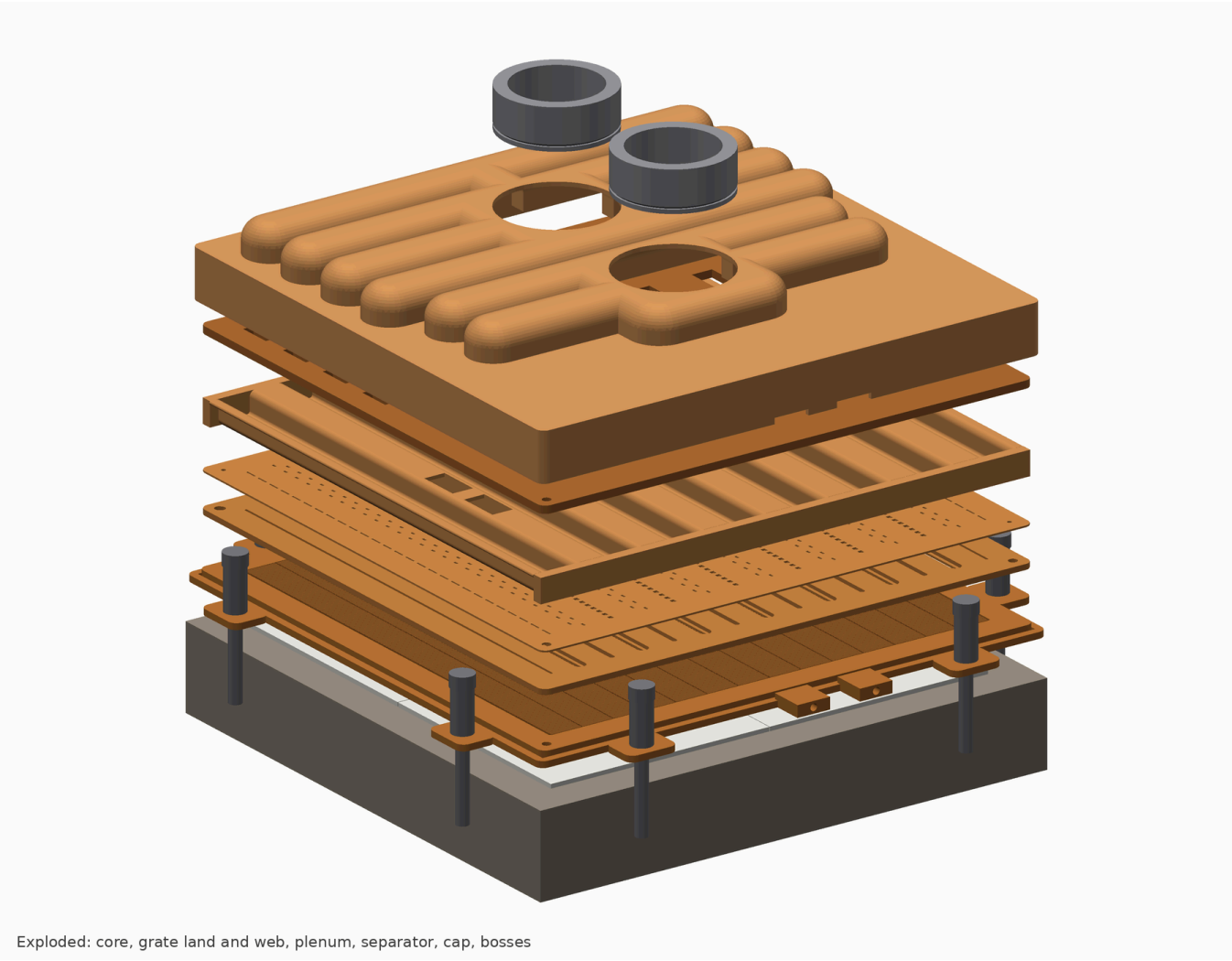


Cross-section close-up: inlet line (orifice web, recess, lip, window, kerf) and outlet line

Coolant arrives in an inlet plenum channel, passes through a 0.5 × 1.05 mm orifice in the web (1.54 m/s, 2.3 kPa), drops through the recess onto the grate lip rather than onto the fins, turns into the 1.2 mm window and the kerf, and enters each fin segment over its full height. It runs 7.7 mm along 0.15 mm gaps at 0.20 m/s, leaves through the kerf and window at the neighbouring outlet line, passes the 0.5 kPa outlet orifice row that balances the two segments either side of a line, and collects in the outlet channels, the four outlet arches and the outlet boss.

Three design rules do most of the work:

1. **Flow length graded by heat flux.** The two-dimensional cell model shows the average case temperature falling with cell length at every flux, while the case ripple is flat between 5 and 8 mm. The 15 kW test plate uses 7.7 mm; a GPU package uses 5 to 6 mm over the dies and 12 mm over memory and margins. This rule turned the architecture's one conceded loss on uniform low-flux lids into parity.
2. **An inertial meter off the braze plane.** The orifices sit in a separate 0.4 mm web, 1.3 mm above the fin-tip joints, so braze fillets cannot reach them, and their sizes (five steps over 1.13:1 in area) cancel the plenum's static-pressure profile at every flow because both scale with velocity squared. Per-orifice flow is uniform within ±2.2 %.
3. **Ties.** Every fin tip is brazed to the grate lands over at least 70 % of its length. The base then sees only the pressure between ties 0.3 mm apart, so it can be 0.85 mm thick with zero pressure bow, which is what lets a pad or liquid metal hold its datasheet resistance over a 108 mm face.

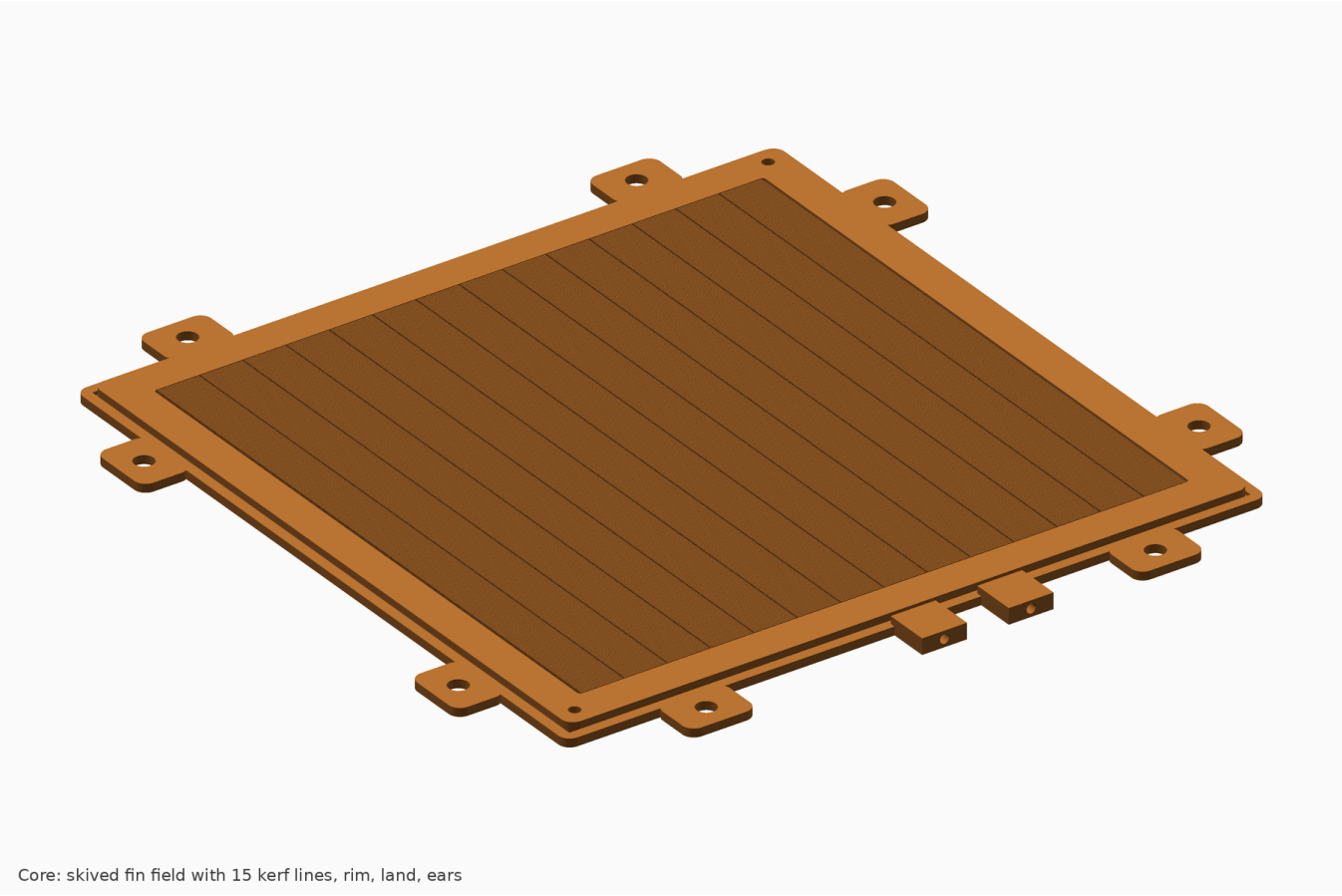


Exploded: core, grate land and web, plenum, separator, cap, bosses

3. Design definition, layer by layer

All dimensions in mm. Axes: X along the fins (across the lines), Y along the lines, Z up from the die face (datum A). The full tables are in `docs/cooling/design/params.json` and the lane documents `DDD-*.md`; the flattened dimension tables per part are in `docs/cooling/cad/drawings/dims_*.csv`.

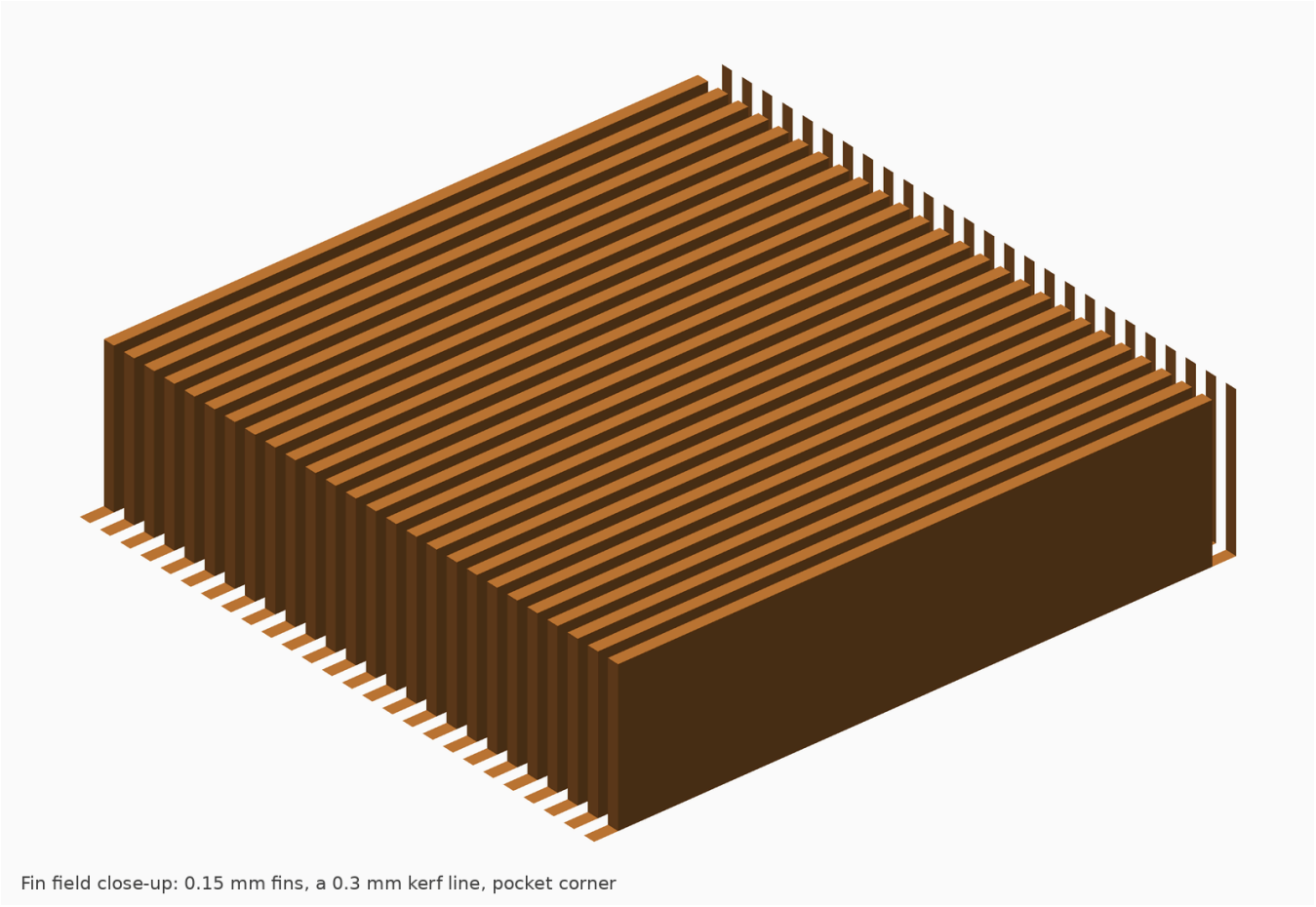
3.1 Core: base, fin field, kerfs, rim, land, ears



Core: skived fin field with 15 kerf lines, rim, land, ears

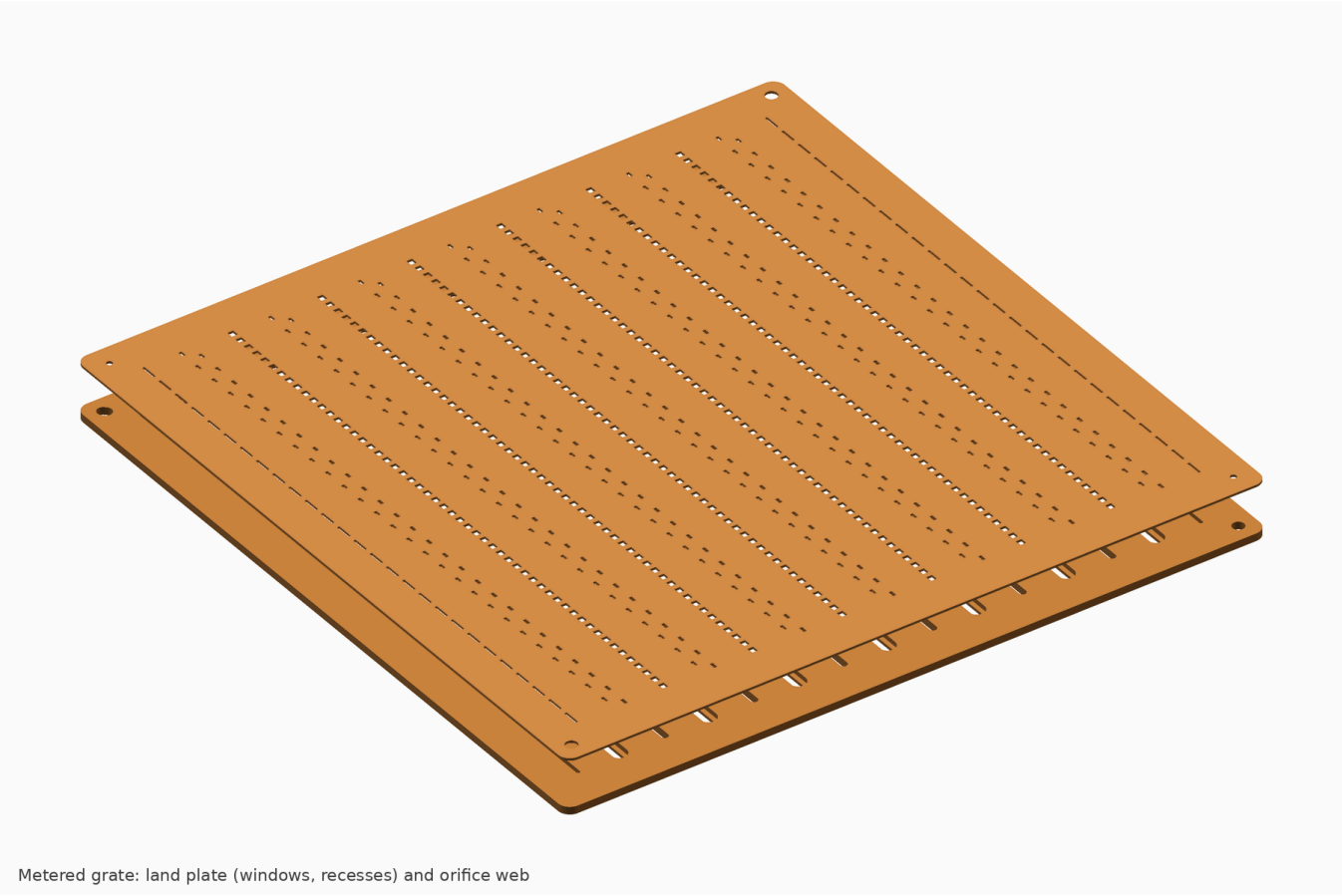
Feature	Value	Tolerance	Basis
Material	OFE copper C10100, H02 plate 4.3 mm, skived at 2.0 mm base, fly-cut to 0.85 after braze	–	DDD-core
Body	125 × 125, corner r 2	±0.1	architecture 5.6
Base under the fin field	0.85 (0.80 min, 1.0 fallback)	+0.05/−0.05	E1 gate on tie detection
Fin field	108 × 108; 360 fins, 0.15 thick, 0.15 gap, 0.30 pitch, tips at z 2.80	t ±0.02, pitch ±0.005, tip plane ±0.02	skiving capability, DDD-core 3
Kerfs	15 lines at x = −54 + 7.7143 k, width 0.30, floor 0.07 above the fin root	w +0.05/−0, x ±0.03	gang saw, potted field
Pocket	108.7 × 108.3, r 0.5, wall top z 2.80; 0.35 mm edge channels for the two edge outlet lines	±0.05	DDD-core
Rim	121.4 outer, top z 2.80, 6.35/6.55 wide: carries the grate and takes the cap skirt lap on its outer wall	±0.05	DDD-core, cap seam
Land	1.8 wide at z 1.5 outside the rim: the one external braze seam	–	architecture 5.1
Ears	8 lobes 12 wide, 7 protrusion, z 1.5, holes 3.4 (4.5 if M4) at (±40, ±66) and (±66, ±40)	±0.1	DDD-core, mech lane

Feature	Value	Tolerance	Basis
Tap bosses	2 on the −y edge at x 0 and 15.43, 8 wide, 1.0 holes at z 1.4 reaching the inlet kerf k = 7 and outlet kerf k = 9	±0.05	fin-field dP signature
Datum pins	2.0 H7 at (±57.5, ±57.5), alumina pins, 1.5 deep	pos ±0.02	DDD-core (grate lane wrote 3.0 at ±57: to be reconciled at G1)
Die face	flatness 0.02 over the fin field, Ra 0.8, parallel to the ear plane 0.03, concave-only form 0–10 µm	–	DDD-core, DDD-mech 7.3



Fin field close-up: 0.15 mm fins, a 0.3 mm kerf line, pocket corner

3.2 Grate: land plate and meter web



Metered grate: land plate (windows, recesses) and orifice web

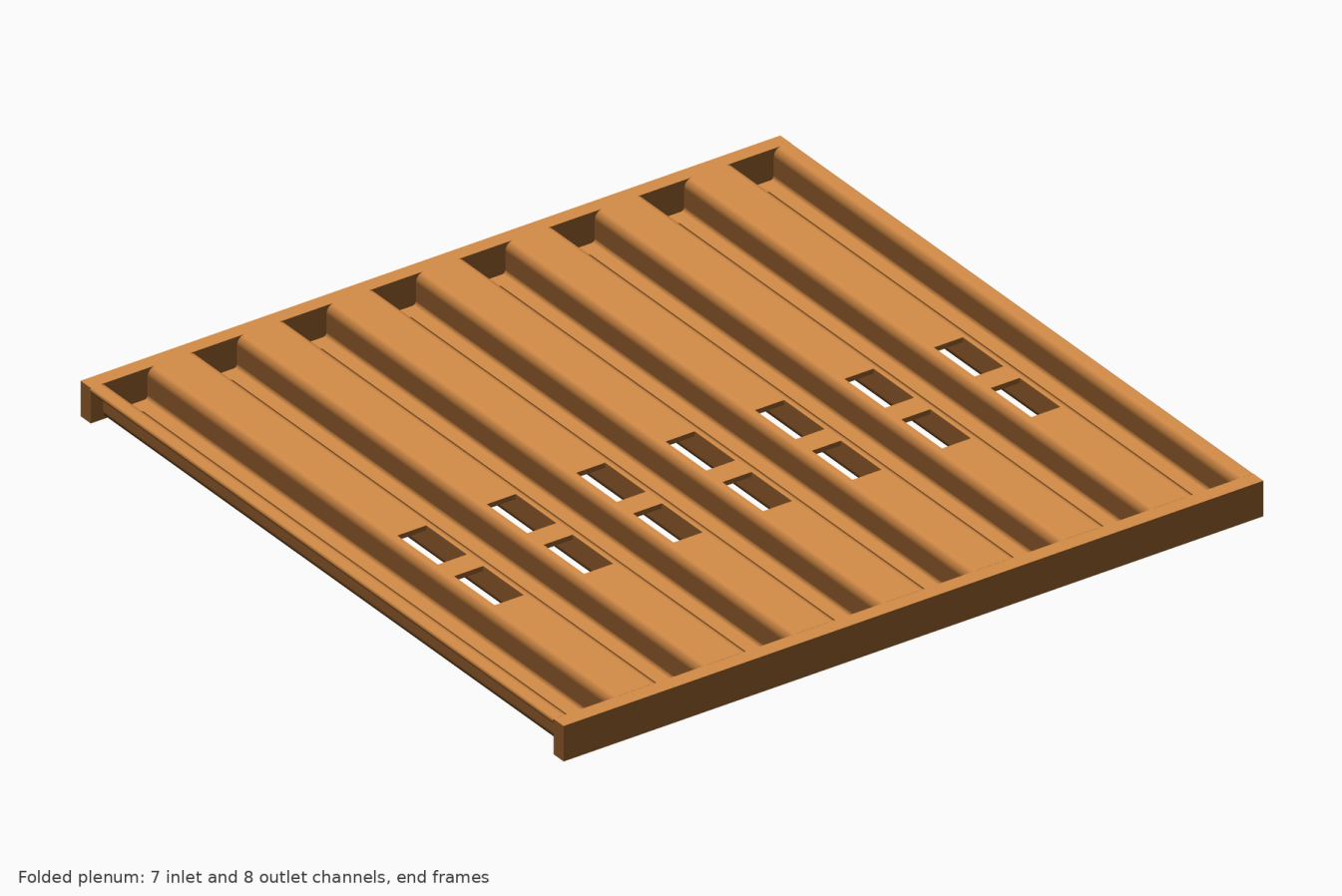
The grate is two laminations brazed in the same cycle, because the recess must open toward the web and no face operation on a single plate can cut that undercut cleanly.

Feature	Value	Tolerance	Basis
Land plate GRT-L	1.30 thick, z 2.80–4.10, OFE H02 strip, both faces skimmed	t ±0.02, flatness 0.02	DDD-grate 3
Windows, 13 interior lines	1.20 wide × 108 long, rounded ends, centred on the line	+0.05/−0	2-D cell model
Windows, 2 edge lines	0.60 wide at x = ±53.7	+0.05/−0	half-line flow
Recess, 7 inlet lines	2.70 wide × 0.80 deep from the top, 109 long, floor 0.50	w +0.05/−0, depth ±0.03	jet lands on the lip; E2 fixes 0.70–0.90
Lip	0.75 wide each side of the window (x 0.60–1.35 from the line)	–	spill margin at the RSS tolerance stack
Lands	continuous, 6.5 wide between lines, braze mask 0.3 from window edges, bond fraction ≥ 0.70 of fin-tip length	–	tie load path
Capillary-break grooves	0.15 × 0.08 laser scribes 0.775 from each window centre (underside) and along the recess edges (top)	±0.02	keeps filler off the window and the orifices
Meter web GRT-W	0.40 thick, z 4.10–4.50, OFE O60/H01 precision strip	t ±0.01	DDD-grate 4

Feature	Value	Tolerance	Basis
Inlet orifices	378: 54 per line at 2.0 pitch, 0.50 wide × 0.99–1.12 long (five size steps), corner r 0.1, alternating ±1.00 from the line so each jet lands on a lip	w, h ±0.01, pos ±0.02	artwork ART-TTV-15-A
Outlet row, interior	324: 54 per line, 1.00 × 1.12 centred on the line	±0.015	0.50 kPa at 18 LPM
Outlet row, edge	54: 27 per edge line, 0.40 × 2.78 at 4 mm pitch	±0.015	0.50 kPa in the 0.6 window
Process	laser pierce from the plenum face, electropolish 3–5 µm, 100 % backlit vision before and after braze; perforating die after artwork freeze	–	DDD-grate 5, DDD-mfg

Meter hydraulics at 18 LPM: 2.31 kPa, mean orifice velocity 1.54 m/s (1.93 m/s at 150 % of rated flow), discharge coefficient 0.72 assumed (0.65 to 0.80 band, measured on coupons in experiment E2), per-orifice flow spread ±2.2 % at every flow from 9 to 36 LPM.

3.3 Plenum and separator



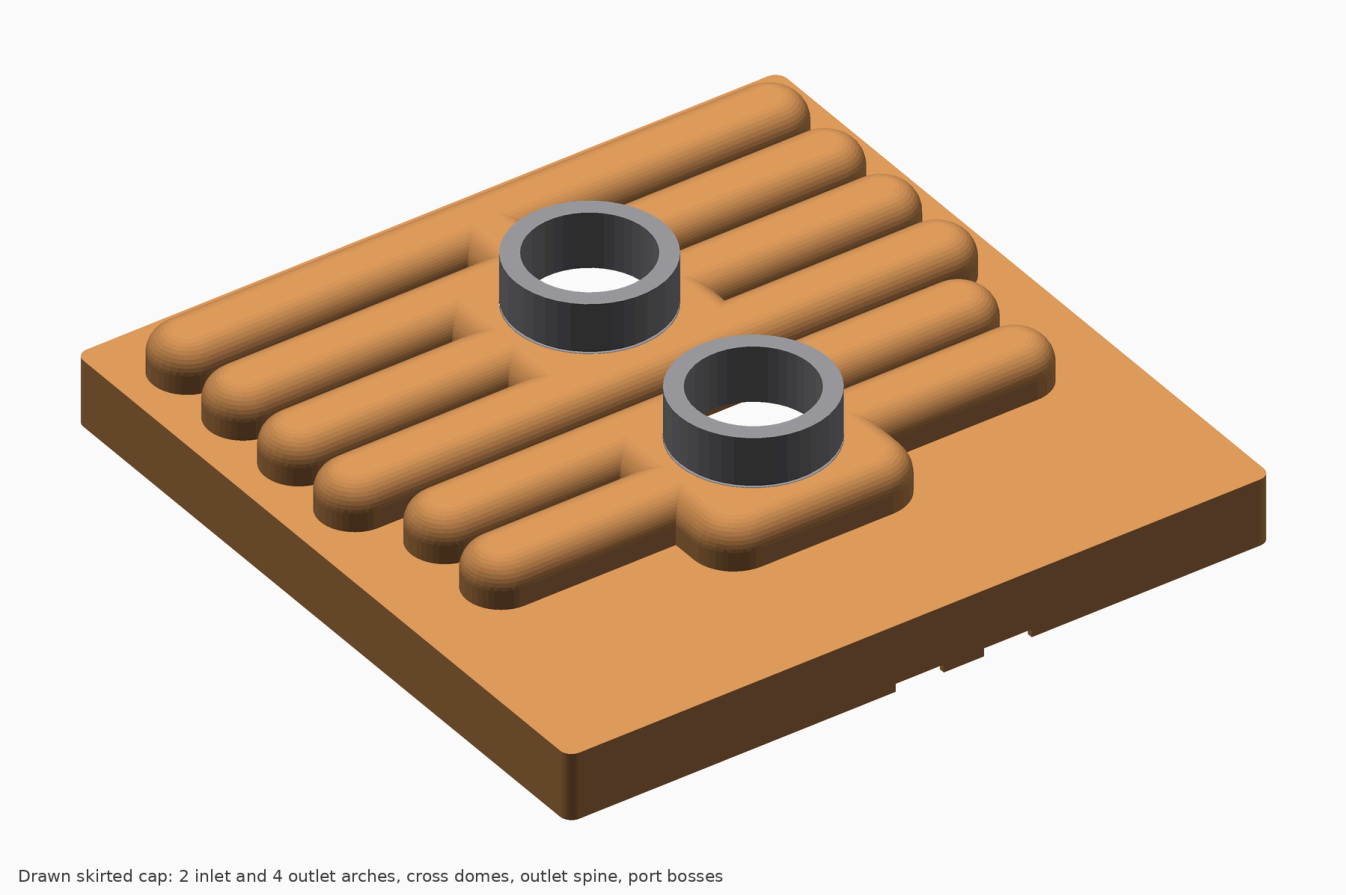
Folded plenum: 7 inlet and 8 outlet channels, end frames

Feature	Value	Tolerance	Basis
Corrugation PLN-C	one 0.80 mm OFE O60 sheet folded into 16 walls, 7 top flats (inlet lines), 8 bottom flats (outlet lines) and 2 outward flanges; z 4.50–10.30	pitch ±0.05, t ±0.03	DDD-plenum 2–3
Channels	15 at 7.7143 pitch, 6.914 wide × 5.00 tall; walls at x = line ± 3.857, 0.80 thick	–	hydraulics 0.65 m/s max
Bends	inside radius 0.6, outside 1.4; K-factor 0.35 (0.31–0.39) fixed on a 3-bend coupon before the die is cut	–	soft copper

Feature	Value	Tolerance	Basis
Developed blank	180.95 × 116.0; fold lines and pierced features in `cad/out/plenum_flat.dxf`	±0.05	DDD-plenum 3.3
Inlet-channel ceiling windows	2 per inlet line, 5.0 × 10.0 at y −21 and −7, feeding up into the separator and the inlet arches	±0.1	plenum solver, win_u ≤ 0.44 m/s
Bottom-flat slots	1.8 × 110 over each outlet row	+0.05/−0	–
Bleed holes	14 × 1.1 mm at z 7.4, alternating y ±55.5, from each inlet dead-end into the neighbouring outlet	±0.02	4 % of plenum flow, no stagnant corner
End frames PLN-F	2 OFE bars 2.5 × 5.78 × 120.7 with 16 sawn slots receiving 1.0 mm wall tabs	slot ±0.03	pitch control at both ends
Separator SEP	1.80 thick, z 10.30–12.10, outline 121.0; 14 inlet windows 5 × 8 at (x_in, −21/−7); 32 outlet windows 5 × 8 at (x_out, +7/+21/+35/+49) with 2.5 wide windows at the edge lines; the plenum's top-flat windows stay 5 × 10	±0.05	plenum lane; mech IR-M3; cap IR-C1 (8 mm keeps a 1.0 margin under a 10 mm arch; 0.54 m/s, 0.29–0.44 kPa)

All plenum and separator joints are internal: a missed joint bypasses a little flow between an inlet and an outlet channel (caught by the infrared flow screen and the pressure-drop test) and can never leak to the outside.

3.4 Cap, headers, bosses and guard



Drawn skirted cap: 2 inlet and 4 outlet arches, cross domes, outlet spine, port bosses

The cap lane (`design/DDD-cap.md`) replaced the provisional 12 × 8 arches. At the 14 mm window pitch a 1.0 mm sheet can only form a brazeable valley between neighbouring arches if the external land is at least 2 mm, which caps the internal width at 10 mm; 7.7 mm of internal height

brings the bar back to the architecture's 20.8 mm. A 26 mm boss needs a 28 mm flat under it, which sets the 35 mm cross domes and moves the bosses to y −20 and +21.

Feature	Value	Tolerance	Basis
Sheet	C10100 OFE 1.00, O60; ≥ 0.70 at the arch ends after draw, 0.92 at crowns and skirt foot	±0.03	DDD-cap 7
Lid	flat on the separator, z 12.10–13.10; load-frame lands at	y	56.5–59.7, x ±60
Arches	6 at 14 mm pitch; internal 10.0 wide × 7.7 high from the separator top, crown corners r 3.5 internal / 4.5 external, top flat 3.0; external 12 × 8.7; valley land 2.0 with r 0.5	±0.15	section 71.7 mm²; DDD-cap 2
Inlet arches	y −21, −7; internal x ±50 (external ±51); half-cup ends r 5 / 6	±0.2	covers the inlet windows with 1.2 margin; mech limit 52
Outlet arches	y +7, +21, +35, +49; internal x ±56 (external ±57)	±0.2	covers the edge windows with 1.05 margin; mech limit 58.8
Header areas per half	inlet 143 mm², outlet 287 mm² (the 2:1 ratio is kept)	–	1.05 / 0.52 m/s at 18 LPM
Inlet cross dome	35 × 35 internal, y −37.5 to −2.5, top flat 28 × 28	±0.2	joins both inlet arches; 2 mm wall to the outlet system at y ±1
Outlet cross dome and spine	35 × 38 internal, y +2 to +40, plus a spine 10 wide to y 54 joining arch 4	±0.2	4.5 LPM at 1.0 m/s in the spine
Height	20.80 at the arch crowns; 29.5 at the bosses	±0.15	12.10 + 1.0 + 7.7 + 1.0
Port pockets and bosses	d26.1 × 1.0 pockets at (0, −20) and (0, +21); OFE bosses OD 26 h9, bore 20 H9, pilot d25.95 × 1.0, z 19.8–29.5; BAg-8 0.6 mm ring at the pilot chamfer	pocket +0.05/−0; position ⌀0.10	inside the mechanical keep-in
Stub seat	counterbore d23 H8 × 8 with a face O-ring groove; 316L stub 20 ID × 23 OD × 30 pressed after the fly-cut and lip-rolled	–	UQD08 adapter or ¾-in barb on the stub
Skirt	inner 121.50, outer 123.4, foot z 1.5, inner corners r 1.0; notched over the two tap bosses; 0.8 mm of the core land exposed outside	inner ±0.03, outer ±0.15	lap gap 0.05 (0.01–0.09) on the rim 121.40
External seam	1.3 mm tall lap × 485 mm perimeter = 627 mm²; one BAg-8 0.5 mm wire ring, 499 mm, 0.98 g, in the outer foot corner; 0.94 MPa shear at design burst	–	the plate's only external seam
Guard GUARD-C	inverted cup d30 × 5.0 of 50 µm absolute 316L weave with a corrugated bottom, 17.4 cm², under the inlet dome at (0, −20); 0.40 kPa clean; installed before braze	–	DDD-cap 6; deviation D-C3
Alternative GUARD-A	pleated 25 cm² element in a d34 × 40 housing on the inlet stub, serviceable, 0.15–0.2 kPa, +12 mm at the port only	–	O-C3, decided with the first customer ICD
Mass	cap 279 g, bosses 38 g, guard 6 g, stubs 32 g	±5 %	blank 188 × 233 × 1.0, 393 g

Pressure design at 3.1 MPa design burst against the 40 MPa allowable (DDD-cap 4): arch top flat 14 MPa, straight walls 27.3, crown corners 10.8, arch-end half domes at 0.70 wall 22.1, dome top ligament carrying the boss ring 37.8, inlet dome walls 27 to 38 and outlet dome walls 28 to 48 depending on how much the arch walls stiffen them (a bound, FEA in E1 as O-C1), and the skirt wall 44.5 MPa at working pressure only if the stack-edge cavity is wetted through two failed internal joints (a secondary leak-before-break boundary, O-C2). Draw sequence: arches, domes and spine are formed first in the flat blank with free feed (channel draw ratio 1.8, arch-end cups 1.97 against a limiting ratio of 2.1), anneal at 450 °C, skirt box draw 125 square by 11.6 deep, restrike to the final radii, lands and pockets, trim to 123.4, pierce the d20 holes and the tap notches.

3.5 Mounting hardware and the load frame

The mechanical lane found that the core's 1.5 mm annealed ears cannot carry spring loads (47 to 195 MPa against a 40 MPa allowable), so every mount uses a load frame that bears on two cap lands at |y| 56.5 to 59.7 mm over the plenum end frames. Eight dia-4 shoulder screws with preset springs give 2,011 ± 80 N on the parity test vehicle (25 psi on 116.6 cm², the pad's datasheet condition), 480 ± 25 N or 880 N on a lidded package (7 or 13 psi), and 0.29 MPa on a bare-die block. The product frame LF-15P is an 85 g 301 stainless stamping; the test frame LF-15T is a fixture. Plate-to-plate clamp repeatability (4 to 5 %) is better than the pad's own pressure sensitivity.

4. Orifice artwork and zoning

The meter web is the only part that changes between packages, and it is laser-pierced, so a new zoning is an artwork file and a two to three week turnaround rather than a die. Two artworks ship with this package:

- `cad/drawings/artwork_prism15_ttv.dxf` (ART-TTV-15-A): the uniform 15 kW test plate, 378 graded inlet orifices and 378 outlet-row holes, computed by `design/grate_artwork.py` from the centre-fed plenum pressure profile so that per-orifice flow is uniform within ±2.2 % from 9 to 36 LPM.
- `cad/drawings/artwork_prism_g_example.dxf`: a zoned example for a four-die GPU package (4 × 26 × 33 mm dies at 350 W/cm² plus 16 HBM stacks) on a 110 × 120 mm field, with 17 flux-graded lines (5.5 mm pitch over the dies, 11 mm over memory and margin) and heat-proportional flow: the die block takes about 88 % of the flow and HBM never sits downstream of a die.

The zoning rule from the architecture: flow length 5 to 6 mm above 250 W/cm², 6 to 8 mm from 100 to 250, 8 to 12 mm from 50 to 100, 12 mm below 50; the die block gets 85 to 90 % of the flow; unheated margin gets slot-end bleeds only.

5. Predicted performance datasheet

Every number in this section is a model prediction from `model/coldplate_model.py` (v3) and the design lanes' own scripts, re-run on the as-designed geometry (0.85 mm base, 1.8 mm separator) by `design/perf_datasheet.py`. The models were cross-checked against each other and calibrated on CoolIT's one published point; they have not been checked against hardware or three-dimensional CFD. Treat each term as ±25 % until experiments E2 and E3 report.

5.1 Term budget at the benchmark point

15 kW, 18 LPM PG25 (k 0.49 W/m·K), 45 °C inlet, 117 cm² heated, eight-thermocouple layout.

Term (K)	Paste	Phase-change pad	Liquid metal (Tier 2)	Note
Coolant caloric rise to the local cell	7.52	7.52	7.52	eps-NTU, NTU 1.16, outlet rise 12.66 K
Thermal interface	8.97	7.05	1.92	0.105 / 0.083 / 0.022 cm²·K/W incl. bond line
Base conduction, 0.85 mm	2.79	2.79	2.79	2.63 at 0.80; 3.29 at 1.0
Convection, fins and base wetted area	10.94	10.94	10.94	h 12.6 kW/m²·K, h_eff 117 kW/m²·K on the footprint
Two-dimensional penalty (cell ends, kerf shadow)	0.87	0.87	0.87	own 2-D conjugate cell model
Average case-to-inlet	**31.1**	**29.2**	**24.1**	CoolIT measured 34.4 (paste)
Case ripple, peak to peak, within a cell	5.39	5.39	5.39	–
Flow hot-spot allowance	1.5	1.5	1.5	weakest segment uses 0.41 K of it
Maximum case-to-inlet	**38.8**	**34.9**	**28.9**	CoolIT measured 44.2 (paste)
Resistance on the footprint, average (cm²·K/W)	0.243	0.228	0.188	CoolIT ≈ 0.27

The as-designed 0.85 mm base costs 0.14 K against the architecture's 0.80 mm; the 1.0 mm fallback costs 0.69 K, and both still beat the benchmark by more than 2.5 K on the average. Channel conditions: 0.198 m/s, Reynolds 44, laminar and fully developed within the first millimetre, fin-channel viscous drop 1.07 kPa.

5.2 Flow sweep in the OCP format

15 kW, 45 °C inlet, PG25, 117 cm². R is case-to-inlet temperature over power; R'' is R times the heated area. The CoolIT-like column is our calibrated model of a split-flow plate, patched to CoolIT's measured point at 18 LPM, and is model-only away from it.

LPM	LPM/kW	Outlet rise K	PCM avg / max K	PCM R'' avg / max	Paste avg / max K	Paste R'' avg / max	LM avg / max K	CoolIT-like paste avg / max K
3.75	0.25	60.8	66.9 / 79.0	0.522 / 0.616	68.8 / 83.0	0.537 / 0.647	61.7 / 73.0	58.8 / 92.6
7.50	0.50	30.4	41.9 / 50.5	0.327 / 0.394	43.8 / 54.4	0.342 / 0.425	36.8 / 44.5	43.5 / 62.1
11.25	0.75	20.3	34.5 / 41.5	0.269 / 0.324	36.4 / 45.4	0.284 / 0.354	29.4 / 35.5	38.4 / 51.9
15.00	1.00	15.2	30.9 / 37.1	0.241 / 0.289	32.9 / 41.0	0.256 / 0.320	25.8 / 31.1	35.7 / 46.8
18.00	1.20	12.7	29.2 / 34.9	0.228 / 0.272	31.1 / 38.8	0.243 / 0.303	24.1 / 28.9	34.4 / 44.2 (measured)
22.50	1.50	10.1	27.4 / 32.7	0.214 / 0.255	29.4 / 36.6	0.229 / 0.285	22.3 / 26.6	33.0 / 41.5

Below 0.75 LPM/kW the 15 kW outlet exceeds 65 °C and the test is run at reduced power; resistance is power-independent to first order. The maximum is lower than the split-flow plate at every flow. The average crosses at about 0.5 LPM/kW: below that the split-flow plate's coarse fins and short caloric path win on the average, which is the conceded low-flow point.

5.3 Capacity, headroom and uniformity

Quantity	Paste	PCM	LM
Load at 85 °C maximum case, 45 °C inlet, 1.2 LPM/kW (thermal)	15.7 kW	18.4 kW	25.4 kW
Rated load (velocity cap: meter 2.0 m/s at 23.4 LPM)	15.7 kW	18.4 kW	19.5 kW
Inlet allowed at 15 kW and 85 °C maximum	46.3 °C	50.2 °C	56 °C

Uniformity level	Rev A at 18 LPM	At 27 LPM	Verification
Along-plenum max/min (graded meter)	1.061	1.07	E2 dye and infrared
Line max/min (diffused port jet)	1.03	1.03	E2
Segment max/min (left/right of a line)	1.06–1.09	1.12	E2 pass ≤ 1.15 at 18, ≤ 1.20 at 27
Weakest segment, extra case temperature	+0.41 K	+0.35 K	inside the 1.5 K allowance

5.4 Pressure drop

PG25 at 45 °C. Plate values are port-to-port with 20 mm bores; 81 % of the drop is inertial, so it scales close to the square of flow.

Element at 18 LPM	kPa
Ports (K 3.5 combined)	1.60
Guard, clean (0.1–0.2 expected, 0.30 booked)	0.30
Headers	0.99
Separator windows	0.46
Plenum channels	0.21
Graded meter	2.30
Windows, kerf turn, miscellaneous	0.16
Outlet orifice row	0.50
Fin channels (viscous)	1.07
Plate, port to port	**7.6**

LPM	% of rated	Plate, port to port (20 mm bore)	With stubs and ½-in barbs	With one UQD08 pair, Cv 4.75 / 3.65	CoolIT-like reconstruction, port to port	Water, port to port
3.75	25	0.57	0.7	1.3 / 1.7	3.5	0.40
7.50	50	1.68	2.2	4.6 / 6.3	7.8	1.31
11.25	75	3.31	4.5	9.9 / 13.7	13.0	2.76
15.00	100	5.48	7.6	17.2 / 23.9	19.1	4.72
18.00	120	**7.60**	10.6	24.5 / 34.2	24.6	6.67
22.50	150	11.41	16.1	37.8 / 52.9	33.9	10.22

LPM	% of rated	Plate, port to port (20 mm bore)	With stubs and ½-in barbs	With one UQD08 pair, Cv 4.75 / 3.65	CoolIT-like reconstruction, port to port	Water, port to port
27.00	180	15.99	22.7	54.1 / 75.8	44.4	14.52
36.00	240	27.46	39.5	95.1 / 133.7	69.3	25.39

External claim: at or below 10 kPa at 18 LPM with 20 mm ports and at or below 30 kPa at 36 LPM. With ½-inch barbs the ratio against the reconstruction falls from 2.7 to about 1.7 because the barbs dominate both plates; with one UQD08 pair (Cv 3.65 to 4.75) the tray-loop figure is 42 to 51 kPa against 56 to 79 kPa. We publish our value against the reconstruction band and never as a measured CoolIT number.

5.5 Inlet temperature, coolant and transients

The case-to-inlet rise moves by less than 0.3 K between 35 and 50 °C inlet (the PG25 viscosity shift changes the fin-channel term only), so the case temperature tracks the inlet one for one. Water is 2 K better than PG25 and 12 % lower in pressure drop at the same flow.

Case (15 kW)	LPM	Pad avg / max K	Case avg / max °C	Paste avg / max K	Case avg / max °C	Outlet °C	Plate dP kPa
PG25, 35 °C inlet	18.0	29.3 / 35.0	64.3 / 70.0	31.2 / 38.9	66.2 / 73.9	47.6	8.05
PG25, 45 °C inlet	15.0	30.9 / 37.1	75.9 / 82.1	32.9 / 41.0	77.9 / 86.0	60.2	5.48
PG25, 45 °C inlet	18.0	29.2 / 34.9	74.2 / 79.9	31.1 / 38.8	76.1 / 83.8	57.7	7.60
PG25, 45 °C inlet	22.5	27.4 / 32.7	72.4 / 77.7	29.4 / 36.6	74.4 / 81.6	55.1	11.41
PG25, 50 °C inlet	18.0	29.1 / 34.8	79.1 / 84.8	31.0 / 38.8	81.0 / 88.8	62.7	7.43
Water, 45 °C inlet	18.0	27.1 / 33.0	72.1 / 78.0	29.1 / 36.9	74.1 / 81.9	57.1	6.67

At 50 °C inlet the paste maximum passes 85 °C; the pad does not. PG30 and EG25 cost 0.5 and 0.1 K against PG25; a 20 °C cold start raises the fin-channel drop to 2.2 kPa and the plate to about 9 kPa.

Transients. The 0.85 mm base holds 0.30 J/K/cm² of near-die heat capacity against 0.69 for a 2 mm base, so under a 1.5× power pulse PRISM rises up to 1.0 K more than a split-flow plate within the first 100 ms. Its absolute peak is still 2.9 to 6 K lower at every pulse length, because it starts lower and settles lower (architecture 6.5: 41.8 vs 44.7 K at 100 ms, 46.4 vs 52.4 K at 1 s on the test vehicle with paste). The plate is not sold as a thermal buffer; the transient KPI is a peak at least 2 K below the incumbent under a 1.5×, 100 ms pulse, measured with a pulsed supply in E3.

5.6 The datasheet block

Item	Phase-change pad (recommended)	Paste	Liquid metal (Tier 2)
R'' average / maximum at 18 LPM (cm²·K/W)	0.228 / 0.272	0.243 / 0.303	0.188 / 0.225

Item	Phase-change pad (recommended)	Paste	Liquid metal (Tier 2)
R average / maximum at 18 LPM, 15 kW (mK/W)	1.95 / 2.33	2.07 / 2.59	1.60 / 1.92
Rise, average / maximum at 15 kW, 18 LPM (K)	29.2 / 34.9	31.1 / 38.8	24.1 / 28.9
Rise at the 15 LPM rated flow (K)	30.9 / 37.1	32.9 / 41.0	25.8 / 31.1
Inlet allowed at 15 kW, 18 LPM, 85 °C maximum case (°C)	50.1	46.2	52.3 (return-limited)
Capacity at 85 °C case, 45 °C inlet, 1.2 LPM/kW (kW)	18.4	15.7	19.5 rated (25.4 thermal)
Plate dP at 15 / 18 / 22.5 / 36 LPM, 20 mm bore (kPa)	5.5 / 7.6 / 11.4 / 27.5	same	same
P10 / P50 / P90 of the average at 18 LPM (K)	26.7 / 29.3 / 32.4	27.9 / 31.2 / 35.4	22.2 / 24.2 / 26.4
P10 / P50 / P90 of the maximum at 18 LPM (K)	32.3 / 35.2 / 38.6	35.6 / 39.3 / 43.7	26.9 / 29.2 / 31.8

The first measurements, in order (DDD-perf section 9): rig repeatability on a purchased incumbent plate; the six-flow pressure drop at three boundaries (the cheapest test and the one that exposes the meter's discharge coefficient); orifice coupons in both flow directions; the fin-field tap reading (1.07 kPa predicted: above 1.3 means filler in kerfs or windows, below 0.9 means missed tip joints); then the average case temperature on the pad at 18 LPM, which is the decision metric for gate G2.

6. Hydraulic and structural margins

6.1 Pressure ratings and load paths

Maximum allowable working pressure 1.03 MPa (150 psi); 100 % proof at 1.55 MPa; design burst at or above 3.1 MPa; sample burst at or above 4.5 MPa; vacuum-stable to −0.9 bar(g). Post-braze annealed OFE copper is designed at 40 MPa (yield 35 to 50).

Load path at 3.1 MPa	Stress (MPa)
Fin ties, bonded fraction 0.70 (0.56 at the gate value)	8.9 (11.1) nominal; 22 (28) local
Folded plenum wall, 0.8 mm at 7.7 mm pitch	29.8
Separator, 1.8 mm over 6.9 mm channels (architecture value at 1.5 mm)	≤ 32.8
Grate cantilever to the outlet window edge	28.6
Cap arch straight walls / crown corners r 3.5 / arch-end half domes at 0.70 wall	27.3 / 10.8 / 22.1
Dome top ligament carrying the boss ring	37.8
Outlet dome walls, foot only / with the arch walls as stiffeners (bound, O-C1)	47.9 / 28.2
Skirt wall at working pressure if the stack-edge cavity is wetted (secondary boundary, D-C2)	44.5 at 1.03 MPa
Perimeter lap seam	safety factor 4.5 to 6.5
Plenum-wall to cap crossings	FEA in E1 (open)

The governing defect is one unbonded land strip spanning kerf to kerf over the 0.85 mm base. It reaches 67 to 99 MPa at proof and would collapse at 1.6 to 2.3 MPa, below design burst, while staying leak-tight at working pressure: a rating problem, not a safety one. Its proof bulge is about 2.2 µm, which the 1 µm confocal scan sees; at a 1.0 mm base it was 0.58 µm and invisible. The thin base is therefore gated on experiment E1 demonstrating per-strip detection; otherwise Rev A ships at 1.0 mm and gives up 0.7 K.

6.2 Velocity and erosion limits

Location	At 18 LPM	Limit	Where it binds
Port bore, 20 mm	0.95 m/s	2 m/s	–
Inlet / outlet arches	0.93 / 0.47 m/s	1.5 m/s	–
Plenum channels	≤ 0.65 m/s	1.0 m/s	–
Meter orifices	1.54 m/s	2.0 m/s (rating), 2.5 m/s (erosion test E4 at 2×)	23.4 LPM: the 19.5 kW rating
Jet on the grate lip	2.2–2.5 m/s local	lip erosion E4, 1,000 h	–
Fin tops under the window	≤ 0.5 m/s (E2 pass)	0.5 m/s	sets recess depth 0.70–0.90
Fin channels	0.20 m/s	–	–

6.3 Braze and mounting loads

The braze fixture puts 0.08 MPa (0.06 to 0.10) on the stack at 795 °C; the folded plenum limits it to 0.10 MPa (wall axial 0.77 MPa, bend corners about 5.6 MPa, fin metal 0.16 MPa against a 0.6 MPa creep limit). Clamp loads in service enter the cap lands at |y| 56.5 to 59.7 mm and pass down the plenum end frames into the rim, never through the fin ties: lands see 3.35 MPa under the 2 kN test clamp and 1.5 MPa under a product clamp against the 40 MPa allowable. Pressure loads (12 kN at working pressure over the fin field) are self-equilibrated by the ties and never reach the clamp budget.

7. Manufacturing

7.1 Route

One vacuum braze cycle joins fourteen copper parts. The routing (`design/DDD-mfg.md` section 2) is, in order: skive the core from a 4.3 mm H02 OFE blank (op 120), pot and gang-saw the fifteen kerfs (130), plane the tip plane to ±20 µm (135), de-wax and deburr (145); mill, slot and groove the land plate and skim both faces (215–230); laser-pierce, electropolish and vision-check the meter web (245); laser-blank, fold and tack the plenum into its end frames (320–330); laser the separator; draw the cap (arches, domes and spine first in the flat blank, anneal, skirt box draw, restrike and trim); place the guard cup under the inlet dome and the boss rings in the pockets; stack on two alumina pins with photo-etched BAg-8 foil preforms (12–15 µm at the fin tips, 25–50 µm elsewhere) in the FIX-15 fixture (520); braze at 795 °C under 10^{-5} mbar with gas quench, 5.5 h cycle, 60 plates per load (610); helium leak test dry (630); infrared flow screen (640); fly-cut the die face to the 0.85 mm base with in-line ultrasonic thickness control (710); flush forward and reverse at 25 LPM (730); proof at 1.55 MPa with a pressurised confocal flatness scan (740); pressure-drop test at 9, 18 and 27 LPM (750); bake, dew-point check and nitrogen pack (760). The 316L stubs are pressed into the boss seats against a face O-ring after the fly-cut and lip-rolled.

7.2 Braze fixture FIX-15

A differential-expansion clamp: graphite platens top and bottom, four TZM tie rods and eight TZM leaf springs. The copper stack grows 0.137 mm more than the rods between 20 and 795 °C, and the springs (5.47 kN/mm total) turn that interference plus a 0.09 mm cold shim into 1,250 N, insensitive to shim error (±0.05 mm gives 0.062 to 0.098 MPa). The base platen has a raised 108 × 108 pad under the fin field so the rim and die-face region bear while the ears float; the top pad is machined to the cap's outer profile with 0.10 mm cold clearance so that it touches the arch crowns and the lid lands together when hot. Boron-nitride stop-off goes on every graphite face and on the skirt foot, never within 2 mm of an orifice, window or kerf. Each fixture is 9.5 kg and 180 × 180 × 95 mm; 60 fit a 900 × 600 × 600 hot zone; life at least 300 cycles. Dead weight was rejected: 0.08 MPa would be 128 kg per plate.

7.3 Inspection plan

Sixteen critical characteristics, each owned by one operation with a reaction plan (`DDD-mfg.md` section 4). The ones that define the product:

CC	Characteristic	Limit	Gauge	Sample
CC-1	Fin thickness and pitch	0.150 ± 0.020 / 0.300 ± 0.005	telecentric vision, 5 fins per plate, Cpk ≥ 1.33	100 %
CC-2	Fin-tip plane and rim coplanarity	flatness ≤ 0.020; coplanar 0.02	laser-line scanner on a 0.5 mm grid	100 %
CC-6	Meter-web hole area and position	area ± 1 % of artwork; position ± 0.020	backlit vision of every hole, checksum	100 % (756 holes)
CC-8	Base thickness after fly-cut	0.85 ± 0.05 (scrap below 0.80)	in-line 10 MHz ultrasonic, 9 points	100 %
CC-9	Die-face flatness, form, roughness	≤ 0.020 over the field; concave 0–10 µm; Ra ≤ 0.8	Fizeau interferometer	100 %
CC-11	Helium leak, dry	≤ 10^{-6} mbar·L/s; Cpk ≥ 1.33 on the log rate	mass spectrometer, accumulation chamber at 10^{-9}	100 %
CC-12	Cleanliness	no particle above 50 µm; ≤ 5 of 25–50 µm (ISO 16232)	membrane count and gravimetry	100 %
CC-13	Proof and pressurised flatness	1.55 MPa, 2 min; no strip bulge ≥ 2.2 µm, no patch ≥ 13 mm	confocal 1 µm on a 0.5 mm grid	100 %; over-proof 1 in 200
CC-14	Pressure drop at 18 LPM (and 9, 27)	7.6 kPa ± 10 %; 27/9 ratio 7.4 ± 0.4; Cpk ≥ 1.33	Coriolis flow, 0.1 % FS transducer, 45 °C bath	100 %
CC-15	Infrared flow screen	no cold line (5–10 mm feature)	640 × 512 LWIR, 30 mK	100 %
Lot	Tie-bond quality	≥ 0.70 bonded per land strip over any 10 mm; no filler within 0.15 of a window; grooves and orifices empty	CT at 20 µm voxel, 1 in 200; sections 1 in 1,000; peel coupon per load	lot
Lot	Burst	≥ 4.5 MPa	hydraulic, 1 in 2,000	lot

Datum scheme (ISO 1101, general ISO 2768-fK): the die face is datum A for the finished plate, the back face A' before the fly-cut; the −x and −y body edges are B and C; the land plate, web and separator register on the two alumina pins D1 (round) and D2 (slot). The full GD&T table per part is in `DDD-mfg.md` section 5.

7.4 Cost

US$ per plate, P50, direct cost of good parts; factory cost = sum ÷ yield × 1.12 overhead. Yield 93.5 % at 100k/yr, 80 % at the 10k/yr ramp. Copper at US$14.6/kg, silver at US$65/oz.

Line	100k/yr	10k/yr
Copper, 1.52 kg of blanks and strip	22.1	25.5
Skive incl. blank mill, pocket, plane	20.5	53.3
Kerf: wax, gang-saw, de-wax, deburr	2.8	7.0
Grate land plate: mill, slot, groove, skim	3.3	9.9
Meter web: laser, electropolish, vision	3.0	4.8
Plenum: pierce, fold, end frames, tack	4.4	14.1
Separator	1.5	3.8
Cap draw and trim, two bosses, boss rings	6.3	18.9
Filler: 5.7 g BAg-8 preforms (4.1 g silver)	14.7	20.6
Vacuum braze, fixture amortised	17.6	38.7
Dies and tooling amortisation	2.0	8.0
Fly-cut, pedestal, in-line gauge	5.6	10.1
Taps, guard, stubs, adapters	8.3	13.3
Hardware: 8 screw sets, product frame, pins, pack	14.0	21.0
Test: flush, helium, proof and scan, dP, IR, bake	11.8	28.3
Direct sum	**137.9**	**277.1**
Factory cost, P50	**165.2**	**388**
P10 / P90	156 / 175	361 / 427

The silver-free route C2 (Cu-Sn transient-liquid-phase belt braze, decided at G1 on E1 coupons) takes the filler line to 4.3 and the braze line to 8.5 for a factory cost of about US$142. Capital per 100k/yr line at the contract manufacturer: 1.4 to 1.6 vacuum furnaces US$2.4–3.2 M; two to three skivers US$1.2–2.1 M; fold tool and progressive die US$0.35 M; draw dies US$0.25 M; fly-cut cell US$0.4 M; test cell US$1.0–1.6 M; total US$5.6–8.0 M.

7.5 Suppliers

Eight process classes, each dual-sourced by month 18: A1 skiving houses (CTX-class machines, 0.15/0.30 × 2.0 mm copper demonstrated on the E1 trial with Cpk ≥ 1.33); B1 precision stamping and machining for the land plate, end frames, separator, bosses and frames (±0.02 on 1.3 to 1.8 mm strip); B2 laser job shops for the meter web (single-mode fibre, ±0.01 to 0.02, 100 % backlit vision); C1 fin-folding houses for the plenum (sequential corrugating, fold tool 8 to 14 weeks); C2 deep-drawing shops for the cap (three-station transfer draw with inter-draw anneal; hydroform for Build A); D1 filler (AWS A5.8 BAg-8 foil, photo-etched preforms, silver hedged 6 to 12 months); D2 filtration (50 and 33 µm absolute square weave, ISO 16889); D3 fasteners and springs (17-4PH H900, 302 preset springs ±10 %); E1 ceramics and graphite (ISO-63 isostatic graphite, TZM to ASTM B386); F the braze house and assembly site (vacuum ≤ 10^{-5} mbar, AMS 2750 class 2, IATF-style quality system, outside Taiwan and not a Group-A cooling supplier, Vietnam or Mexico); G our own test equipment (helium, confocal, dP, IR, flush, bake; US$1.0–1.6 M per line).

7.6 First-article qualification

Step	What	When
F1	Material receipt: mill certificates (oxygen ≤ 5 ppm, conductivity, grain), strip thickness maps	M1
F2	Core process trial: skive 6 blanks, pot, saw and plane 4, section 2; Cpk estimate on CC-1	M1–2
F3	Grate and plenum coupons: recess and lip ±0.03; three-bend fold coupon fixes the K-factor	M1–2
F4	Braze coupons (E1): 24 stacks across routes A, B, C, fixture pressure 0.04 / 0.08 / 0.12 MPa, tip plane ±10 / ±20 µm, potted vs unpotted kerfs; tie bond ≥ 0.70, grooves empty, per-strip bulge detection shown	M2–3
F5	FIX-15 commissioning: 3 fixtures, load cells cold and witness thermocouples at 795 °C; 1,250 ± 250 N hot	M2
F6	Build A: 12 full plates on laser, CNC, hydroform and wire-EDM parts; 2 to CT and sections, 1 burst, 1 over-proof, 8 to the water table and the thermal test vehicle	M3–4
F7	Dimensional first-article report: ballooned drawings, CMM on 3 of each part	M4
F8	E2 hydraulics: dP 7.6 ± 10 % at 18 LPM; segment max/min ≤ 1.15; discharge coefficient both directions	M4
F9	G1 freeze: artwork, recess and lip, base 0.8 vs 1.0, route A vs C2; release production tooling	M3–5
F10	Build B pilot, 300–500 plates on production tooling; 30-plate capability study; PPAP-style package	M12–18
F11	Second-source first articles at the second skiving and braze houses	M18
F12	Reliability and compatibility samples from Build B	M12–24

8. Mechanical integration

8.1 Load path and hardware

The core's 1.5 mm annealed ears cannot carry spring loads (47 to 195 MPa against 40 MPa allowable), so a load frame bears on two flat cap lands at |y| 56.5 to 59.7 mm, x ±60, z 13.10 ± 0.10, directly over the plenum end frames. The product frame LF-15P is an 85 g 301 stainless stamping with hook tabs that ships with the field-replaceable unit; the test frame LF-15T is a fixture (aluminium 0.46 kg or steel 1.34 kg). Eight dia-4 captive shoulder screws (17-4PH H900) with preset 302 springs, stop sleeves and e-clips give a kit per package class: SPR-60 (480 N total, 7 psi on a 100 cm² lid) or SPR-110 (880 N, 13 psi); the DIE-TTV kit gives 2,011 ± 80 N on the test vehicle. The ear holes are dia 4.5 +0.10/−0 by default (core lane listed 3.4 with 4.5 as the M4 option; mechanical lane request IR-M1).

8.2 Retention force budget

Interface	Pressure window	Area cm²	Force N	Per screw N	Kit
TTV-A, four 54 × 54 AlN heaters, pad datasheet condition	25 psi	116.6	2,011	251	DIE-TTV
TTV-100 hard mode, four 50 × 50 copper blocks	25 psi	100.0	1,724	216	DIE-TTV, carrier stop raised 1.04
Paste comparison runs	20 psi	116.6	1,609	201	DIE-TTV with 1.5 mm shim

Interface	Pressure window	Area cm²	Force N	Per screw N	Kit
Lidded 100 cm² package, pad	4–12 psi	100	276–827	35–103	SPR-60 or SPR-110
Lidded package, liquid metal	5–15 psi	100	345–1,034	–	SPR-60
Bare-die two-die package, 16.6 cm² die block	0.15–0.35 MPa	16.6	249–581	–	SPR-60 (0.29 MPa on the die)
Bare-die four-die package, 33 cm²	0.15–0.35 MPa	33	495–1,155	–	SPR-60 or SPR-110

Plate-to-plate clamp repeatability is 4 to 5 %, better than the pad's own pressure sensitivity.

8.3 Bow and the die-face form rule

Under the 2 kN test clamp the plate bows 22.6 µm concave (two-side bearing, closed form); the hot base against the cooler stack bows 18.7 µm convex in operation, and the two nearly cancel. The rule that follows: the die face is fly-cut concave by 0 to 10 µm (test plates to 5 µm) and never convex, so corners never lift hot. Differential in-plane expansion over 108 mm from 20 to 85 °C is 0.12 mm against a ceramic or silicon package and is taken by the compliant interface. Verification remains an FEA of frame and plate at 0.48, 0.88 and 2.0 kN and a heated, pressurised flatness scan on five plates (open item O-M2).

8.4 Test vehicles and thermocouples

TTV-A carries four 54 × 54 mm aluminium-nitride heaters at 54.3 mm pitch (116.6 cm², the parity vehicle, up to 18 kW); TTV-100 carries four 50 × 50 mm copper-block cartridge heaters (100 cm², the hard-mode vehicle, 30 kW with TTV-A2); TTV-25 reproduces CoolIT's 4 kW single-die point; TTV-D is a copper-neck vehicle to 400 W/cm². Eight thermocouples replicate CoolIT's inferred two-per-heater layout (centre and outer edge of each heater), to be confirmed against a purchased CoolIT vehicle in month 1 (O-M7). Heat capture is required at 98 % or better and re-mount repeatability within 1 K before any comparison run is counted.

8.5 Variants

- **HBM pedestal (Tier 3 bare-die SKUs, PRISM-G-D).** A stepped pedestal 0 to 0.40 mm, signed, ±0.05 (±0.02 for steps ≤ 0.2), cut in the same fly-cut as the die face; a raised root costs about 0.7 K per 0.25 mm on the memory.
- **PRISM-G envelope (GPU class).** 127 × 137 mm body, 110 × 120 mm field, 17 flux-graded lines (5.5 mm over the dies, 11 mm over memory), mass 1.03 to 1.15 kg by scaling Rev A (about 1.05 kg with a 316L separator); horizontal dia-10 stubs at z 12 for 1U stacks are the cap lane's open item.
- **Handling.** Carrier pocket, poka-yoke on the asymmetric port positions, handling cover over the die face, N2 pack at ≤ −40 °C dew point.

8.6 Mass budget

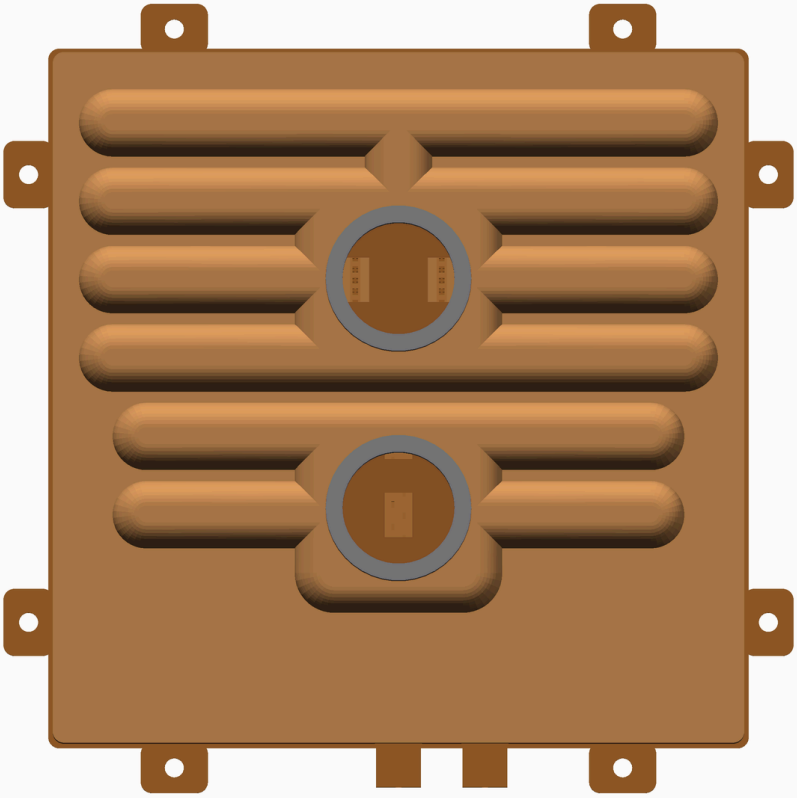
Part	Lane g	CAD g
Core (base, fins, rim, ears, tap bosses)	282.0	285.1
Grate land plate + meter web	201.0	190.9
Plenum corrugation + 2 end frames	161.6	167.4

Part	Lane g	CAD g
Separator, 1.8 mm copper, 46 windows 5 × 8	173.7	208.4
Cap, drawn, skirted (cap lane)	279	233.6
Port bosses (2)	38	37.4
Guard GUARD-C, 8 screw sets	6 + 80.4	3.3 + 24.7 (shell and screws only)
Stubs (2), backplate, standoffs	119.1	not modelled
Plate on the architecture boundary	**1,307 (cap lane)**	**1,151 + stubs, backplate and standoffs**
+ product load frame LF-15P	85 → 1,363	–

The architecture budget is 1,240 g with a 1,400 g SKU limit. On the cap lane's basis the plate is 1,307 g, 67 g over budget and 1,392 g with the product frame. The CAD cap solid is 233.6 g against the lane's 279 g: the lane's developed-area sum counts the lid under the arch and dome footprints twice (about 64 g), so the corrected lane total is about 1,262 g. A 0.6 mm 316L separator (−106 g) recovers the budget either way; a thinner cap sheet is ruled out by the 0.70 draw-thinning floor at the arch ends. Decision at G1.

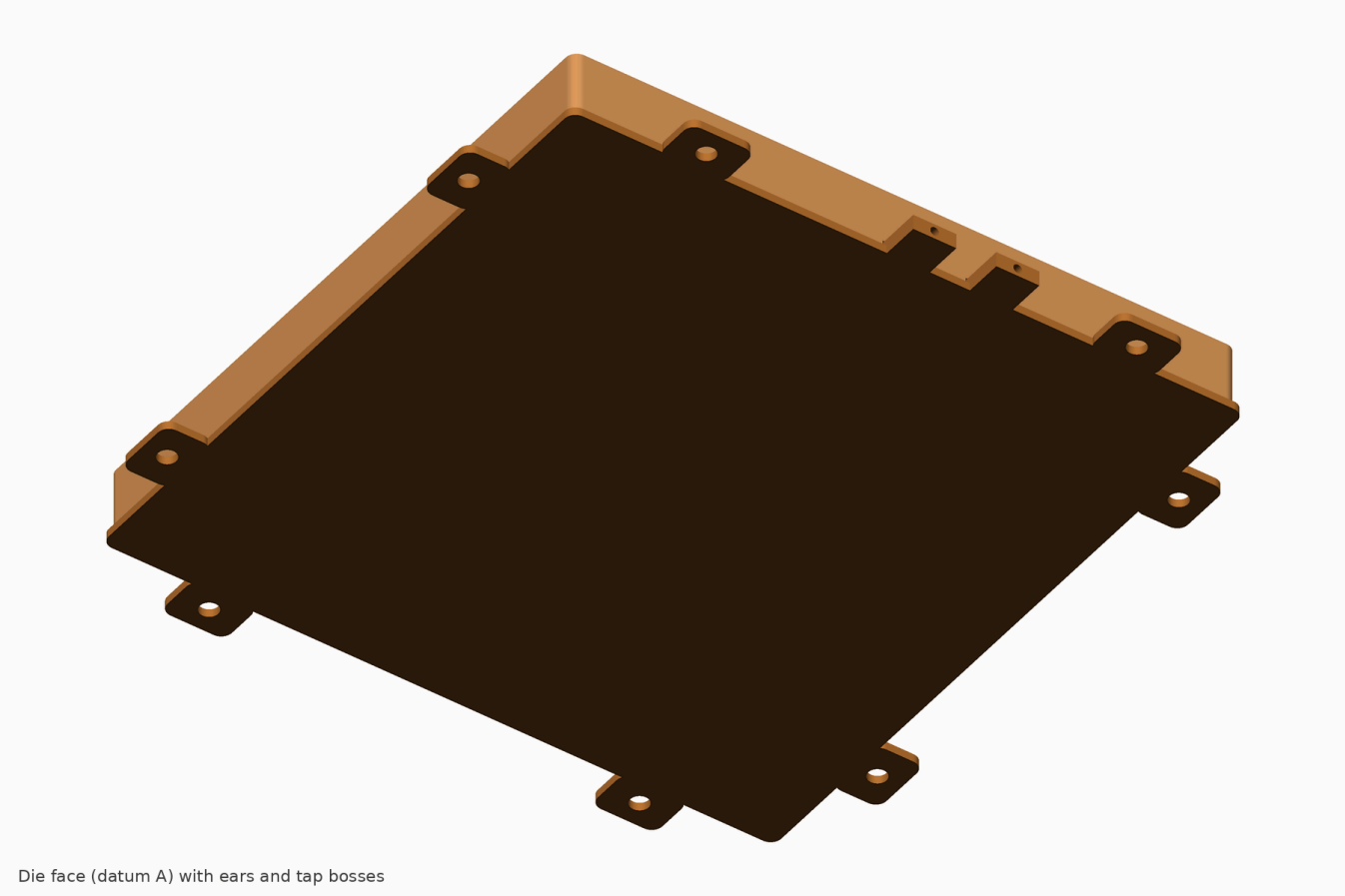
9. CAD package and checks

The CAD is parametric: fourteen solids built by CadQuery 2.8 on OpenCascade from the one parameter file `design/params.json`, so a changed dimension regenerates every part, the assembly, the interference check, the renders, the drawings and the mass table. Every part is modelled with its real features: 360 skived fins with fifteen kerf lines, 756 orifices in the meter web, windows and recesses in the land plate, the folded plenum with its bleed holes, the drawn cap with six arches and two cross domes, bosses, ears, tap bosses and datum pins.



Top view: 2 inlet arches, 4 outlet arches, 20 mm port bosses

Deliverable	Files
STEP AP214 per part and the full assembly (one node per part), xz-compressed	`cad/out/*.step.xz` (core 7.6 MB compressed, 89 MB expanded; assembly 8.6 MB)
Tessellated meshes per part and the four section solids, 0.05 mm chord	`cad/out/*.stl`
Build reports: validity, solid count, volume, mass, bounding box; interference and mass checks	`cad/out/*.json`, `cad/out/assembly_report.json`
Developed flat pattern of the plenum with fold lines and pierced features	`cad/out/plenum_flat.dxf`
Orthographic and isometric views of each copper part and the assembly	`cad/drawings/*.svg`
Orifice artworks for laser piercing, separator window pattern, plenum blank	`cad/drawings/*.dxf`
Dimension tables per part for the drafter	`cad/drawings/dims_*.csv`
Report figures	`cad/renders/*.png`
Source	`cad/prism_cad.py`, `cad/parts/*.py`, `cad/assembly.py`, `cad/render_views.py`, `cad/drawings.py`, `cad/README.md`

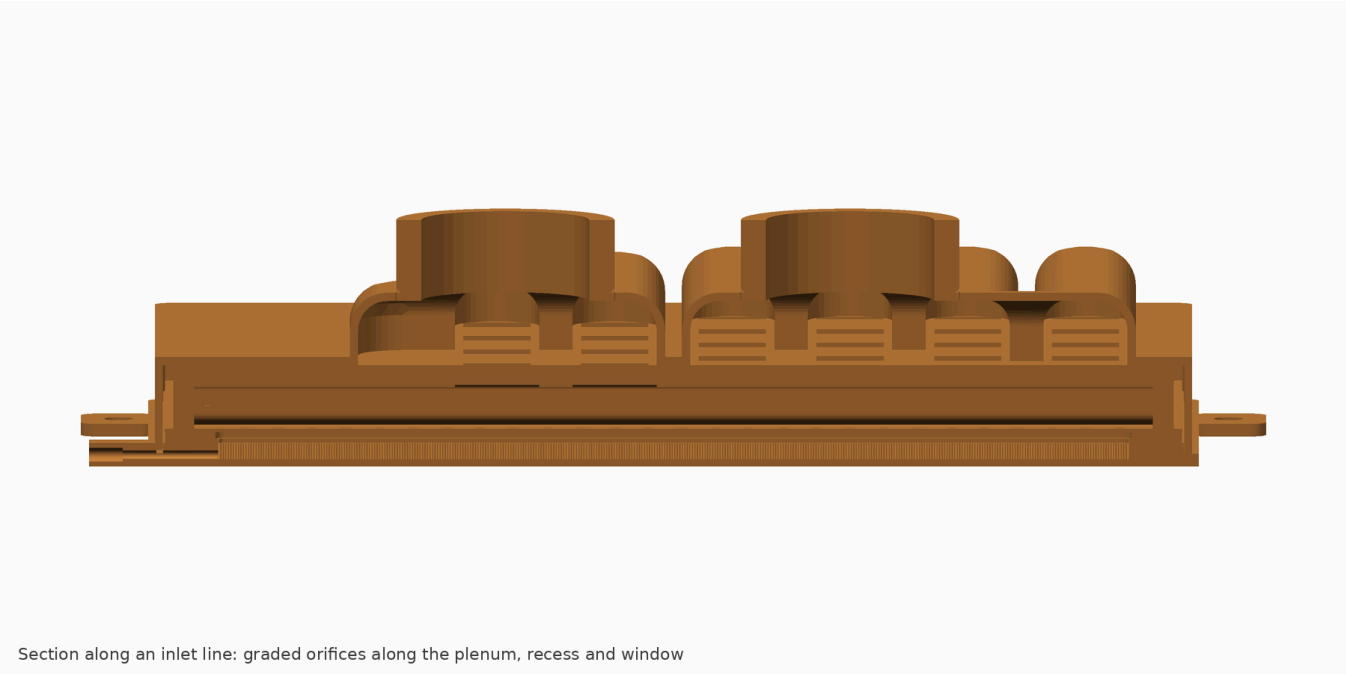


Die face (datum A) with ears and tap bosses

Checks on the full-resolution build, regenerated with the cap lane's geometry. All fourteen solids are valid single bodies. Twenty pairs were checked for interference (the stacked copper layers, the cap against the core and against every part inside its skirt, the bosses and the guard against the cap and each other) and all return zero volume. The 121.5 mm skirt sits on the 121.4 mm rim lap face with a 0.05 mm radial gap and clears the separator and grate (121.0) by 0.25 mm, the plenum flange tips by 0.24 mm and the end frames by 0.40 mm; it is notched over

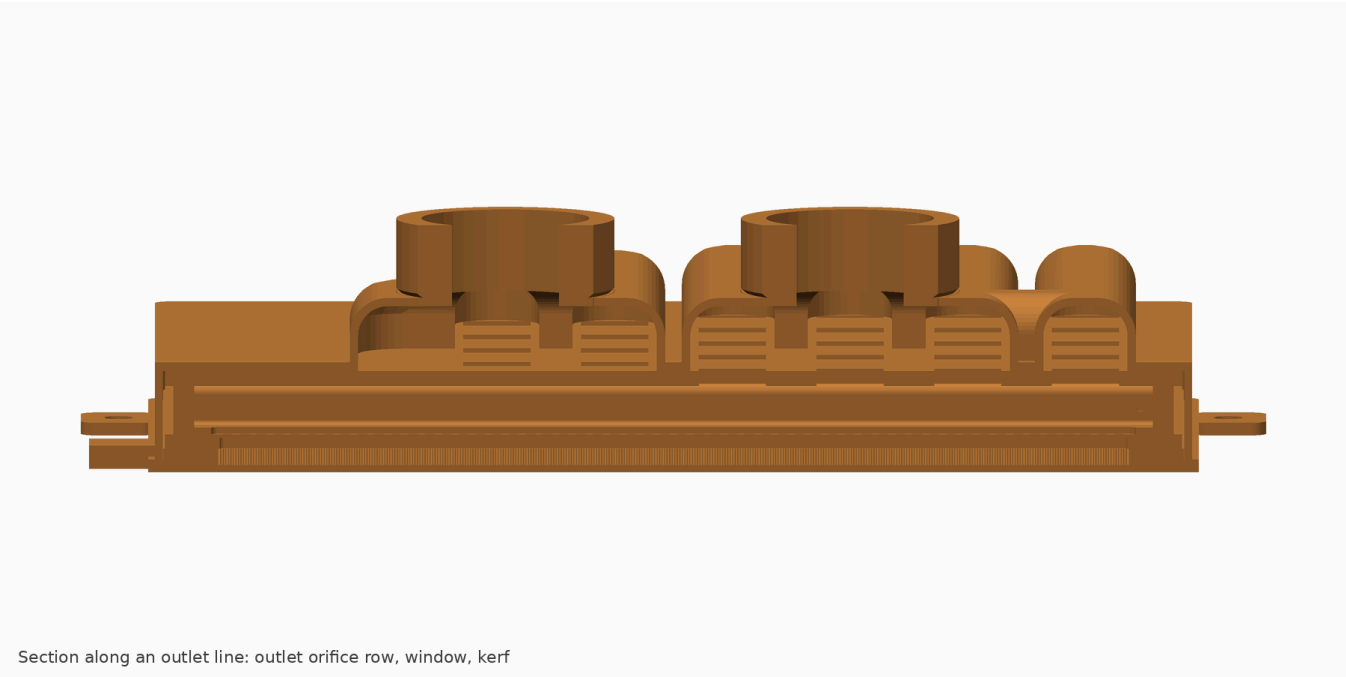
the two tap bosses. Height 20.8 mm at the arch and dome crowns, 29.5 mm to the top of the port bosses. Plate mass 1,151 g without stubs, frame or test vehicle (table in section 8.6). Stack: die face 0 | base to 0.85 | fins to 2.80 | land plate to 4.10 | meter web to 4.50 | plenum to 10.30 | separator to 12.10 | cap lid to 13.10, crowns to 20.80 (internal 19.80) | bosses to 29.5.

Two findings from the regeneration. First, with half-cup arch ends the straight part of each arch stops at |x| 45 (inlet) and 51 (outlet) while the outermost separator windows reach |x| 48.8 and 54.95, so the outer corners of the four inlet windows at x ±46.3 (1.3 % of their area) and the eight edge outlet windows at x ±53.7 (4.0 %) sit under the arch end wall; the mechanical limits (52 / 58.8) leave room to lengthen the arches by about 1 mm, which the cap and plenum lanes decide at G1 (O-C6). Second, the cap lane's 279 g sums the full lid area and then adds the arch and dome developed areas, counting the lid under their footprints twice; the CAD solid at a uniform 1.0 mm wall is 233.6 g.



Section along an inlet line: graded orifices along the plenum, recess and window

Known simplifications. Kerfs are modelled at full depth over the whole fin length (the saw run-out ramp beyond |y| 48.1 mm is not). The 0.15 × 0.08 mm capillary-break scribes, the 0.1 mm orifice corner radii, the plenum wall end tabs and end-frame comb slots are not modelled; orifice areas are within 2 % of the artwork. The guard is a 0.1 mm shell without its corrugated bottom, feet and hook tabs, hung from the dome crown 0.7 mm higher than the lane's feet place it; the cap walls are a uniform 1.0 mm (the drawn part thins to 0.92 at the crowns and 0.70 at the arch ends); the stub-seat counterbores, O-ring grooves and the 316L stubs are not modelled. The outline is 121.0 mm (confirmed by the cap lane) and the pins 2.0 mm at ±57.5 (core lane) where the grate lane wrote 3.0 at ±57; the pins go to G1. The screws and the load frame follow the mechanical lane.



Section along an outlet line: outlet orifice row, window, kerf

10. Verification record, concessions and risks

10.1 How the design was stress-tested

The architecture passed five review rounds before this design (`04-verification.md`): a physics check that corrected the model (Darcy friction, isothermal caloric term, PG25 conductivity 0.49 W/m·K, the plate's own two-dimensional penalty rather than CoolIT's calibration-absorbed one); a design panel of eight proposals with independent judges; an adversarial verification across physics, hydraulics, manufacturing, cost and market, and IP; a completeness critic; and a reproducibility audit in which all six final scripts re-ran byte for byte. The largest changes it forced: the 18-line layout became 15 lines with edge outlets; the stamped laminations became a folded plenum and a drawn cap; the doubled outlet header and the outlet orifice row cut segment imbalance from 1.36 to 1.06; jets moved from fin tips to the grate lip; cost was re-based from US$115 to US$165; the US launch was tied to a freedom-to-operate opinion.

The design lanes then re-derived every number on the dimensioned geometry: the core, grate, plenum, mechanical and manufacturing lanes each carry a deviations table against the architecture and an open-items table with the test that closes each item. The performance lane re-ran the benchmark on the as-designed 0.85 mm base and found the architecture's numbers within 0.14 K.

10.2 What the design does not win, stated plainly

Concession	Size	Why it stands
Price per plate	US$165 vs about US$126 at 100k/yr (+31 %)	fine skiving, two-lamination grate, silver filler, 100 % test; parity on cost per kW of capacity; US$142 on the C2 route
No field record	–	first hardware is month 3; 22-plate endurance is months 5–11
Fouling life at equal filtration	0.15 mm gaps vs 0.2–0.3	ships with a mandatory 33 µm absolute tray filter (decision D6); dose claim only until E4

Concession	Size	Why it stands
Average on a 100 cm² footprint, paste	parity	the fin-field advantage shrinks when CoolIT's heaters are small; we test both vehicles
Average at or below 0.5 LPM/kW	loss	the long caloric path of a split-flow plate wins on the average at low flow; maximum still wins
CoolIT's 4 kW single-die point	loss with the drop-in core	the metered grate pays its 2.3 kPa and the fin field is oversized for a 50 × 50 die
350 W/cm² at 85 °C case	miss (about 98 °C drop-in, 88 °C filtration-gated grade)	about 16 K better than a split-flow model; a bare-die Tier 3 problem, not the 15 kW test
Thermal buffering of power steps	40 % less copper over the die	not sold as a buffer; the loop handles transients

10.3 Risk register (ranked)

#	Risk	Likelihood / impact	Early warning	Fallback
R1	Tie braze: voids, wicking, kerf damage; per-strip detection fails at the 0.85 mm base	medium / critical	E1 at M3	routes B, C, D; base 1.0 (+0.7 K); no-kerf core outside the US only
R2	Segment uniformity worse than 1.15	low–medium / +1–2 K on the maximum	E2 at M3	re-artwork the outlet row (+0.5 kPa)
R3	Meter discharge CFD shows fin-top velocity above 0.5 m/s	medium / erosion, IP	CFD at M2	wider lip, lower orifice velocity
R4	Benchmark ambiguity used against us	high / credibility	reviewers	two vehicles, R'' per area, eight-TC replication, purchased plates, standing offer to test any CoolIT plate
R5	OCP's recommended interface material is a paste	medium / average margin falls	read OCP §5.2 in M1	lead with maximum, uniformity, dP, capacity
R6	Fouling or kerf impaction	low–medium / life claim	E4	dose claim only; mandatory cartridge
R7	Customer mandates a nickel wetted path	medium / socket access	statement-of-work reading	route C with pre-plated nickel
R8	IP: Split-Flow, TW I606224, Microfabrica	high / US timing	FTO at M4	non-US first; US from Sep 2028; indemnity insurance
R9	Cost above US$175 at 100k	medium / channel	RFQs at M1	C2 route; per-SKU tooling limited to the meter web
R10	TTV heaters crack at 150 W/cm²	medium / schedule	commissioning M3–4	copper-block cartridge vehicles
R11	Supplier capacity (furnaces, skivers)	medium / schedule	RFQs	two sources each; long-lead orders at M1
R12	Micro-channel lids displace lidded plates at NVIDIA	medium–high / rank A2	customer signals	D7 lid-insert study; G2 pivot decision
R13	Section 232 classification	medium / landed cost +US$50–110	CBP ruling	site decision tree

#	Risk	Likelihood / impact	Early warning	Fallback
R14	CoolIT publishes a better plate in its own OCP format	medium / marketing	market	R+ and liquid-metal data in reserve; transparency

10.4 What has not been done

No hardware, no three-dimensional CFD, no reading of the OCP specification text itself (it could not be fetched from this environment; its recommended interface material and case definition matter), no legal opinion. The first three months of the programme exist to close exactly these gaps.

11. Innovation register

The elements below are what a competitor must copy to match the plate, with the protection route the architecture assigns to each.

#	Element	What it does	Why it is new	Protection
1	Tied fin tips: every fin brazed to the grate lands over ≥ 70 % of its length	base sees pressure only between ties 0.3 mm apart; 0.85 mm base with zero pressure bow; MAWP carried by the ties	skived plates are open-topped; the few fin-to-lid braze proposals (Nidec) have no manifold on the tie	file (filing 1), narrow with numeric limits
2	Kerfed skived field with a metered grate	7.7 mm flow length at 0.15 mm gaps without the dP of a long split-flow path; every segment starts at inlet temperature	manifold microchannels exist (Cooligy, JetCool); none is a brazed skived field with kerf-fed cells and a stepped grate	filing 1
3	Inertial meter off the braze plane, graded to the plenum profile	±2.2 % per-orifice uniformity at every flow because both the meter drop and the plenum profile scale with velocity squared; braze cannot reach the orifices	microjet plates put the orifices in the braze plane; grading is usually viscous (flow-dependent)	filing 1; artwork tool as trade secret
4	Jets landing on a grate lip over a recess, not on the fins	fin-top velocity ≤ 0.5 m/s, no erosion, clears the Microfabrica family	the usual impinging-jet-on-fin arrangement is both an erosion risk and an IP risk	filing 1; chart after CFD
5	Flux-graded flow length (5–6 mm above 250 W/cm², 12 mm below 50)	zoning by cell length rather than by flow alone; die block takes 85–90 % of the flow with HBM never downstream of a die	partly anticipated (Cooligy, Toyota, HUST)	defensive publication
6	Doubled outlet header plus a 0.5 kPa outlet orifice row	balances the two fin segments either side of each line (1.36 to 1.06) at no height cost	segment-level imbalance is unaddressed in manifold plates	filing 1
7	Folded 0.8 mm plenum with bleed holes and end frames	15 channels at 7.7 mm pitch from one sheet; no stamped webs; internal joints cannot leak outward	stamped laminations were found unmakeable at these webs	know-how
8	Drawn skirted cap with one external seam	the only joint that can leak to the outside is a lap seam with a safety factor of 4.5–6.5	most plates have four to six external seams	know-how
9	Fin-field pressure-drop taps	a fouling signature read at the fins, not at the ports; service on evidence rather than on hours	plates have no in-core telemetry	publish via OCP

#	Element	What it does	Why it is new	Protection
10	Capillary-break grooves and masked lands	keep filler out of windows and orifices in a single braze cycle	–	trade secret
11	Differential-expansion braze fixture (TZM springs, graphite platens)	0.08 MPa on the stack insensitive to shim error; 60 plates per load	dead-weight fixtures are impractical at this pressure	trade secret
12	Per-strip bulge detection at proof (confocal, 1 µm)	finds the one unbonded land strip that passes leak and proof	the governing defect of any tied plate; no published control	trade secret
13	Filtration ladder: 50 µm guard in the boss plus a 33 µm tray cartridge	lets the fine field be sold with a stated life	–	publish via OCP
14	Artwork-per-SKU zoning	a new package is a laser file and two to three weeks, not a die	–	trade secret (tool)
15	Transparent test method: two vehicles, three dP boundaries, independent repeat	makes the comparison un-attackable	–	defensive publication after clearing US 12,446,195

The innovation lane (`design/DDD-innovation.md`) carries each of the ten design-level elements as a full card: the problem it solves, the mechanism with its dimensions, the quantified benefit with the script it comes from, the closest prior art and the difference, a novelty confidence (high, medium, low, none) under the freedom-to-operate caveat, and the test that proves it. Its headline judgement: the strongest patentable element is the stepped recess with lip landing and the off-braze-plane meter (medium confidence, the core of filing 1); the short flux-graded flow length and the tie itself are low-confidence as novelty and are positioned as a proven subset done well; the thin-base rating logic, the per-strip screen and the artwork solver are trade secrets. It also lists what is deliberately not claimed (skived copper, vacuum brazing, split manifolds as such, orifice-compensated headers, PCM and liquid-metal interfaces) so that marketing and the filing never overreach.

Stretch innovations, not in Rev A (DDD-innovation section 4), carried as options for a Rev B once the braze is proven:

- **S1. Chevron-staggered lines.** Offsetting adjacent lines by half a segment along y would put the hot end of each cell beside the cold end of its neighbour 0.3 mm away, cutting the conduction length that smooths the kerf-to-kerf ripple from the 3.7 mm half-pitch to about 2 mm. Ripple smoothing scales with length squared, so the 5.4 K peak-to-peak ripple could halve, worth about 1.3 K on the maximum and nothing on the average. Cost: gang saws cut straight lines, so the kerfs become wire-EDM or laser work and the plenum channels become chevrons.
- **S2. Viscosity-driven self-balancing for zoned SKUs.** The fin drop is viscous, and PG25 viscosity falls about 30 % between 35 and 50 °C, so a segment under a hotter zone would draw 10 to 15 % more flow by itself if part of the inertial meter budget were moved into a laminar element. The price is that the plenum-profile sensitivity the velocity-squared meter cancels comes back in proportion.
- **S3. A 0.60 mm base.** Worth about 0.8 K on every interface. The ties make it structurally free, but one unbonded strip would then collapse at or below working pressure, so it needs a

second independent 100 % tie screen (lock-in thermography on the as-brazed face) before it can be rated. It is the natural follow-on once E1 has shown what the confocal scan can see.

12. Applications

Rank	Application	Variant	Entry
A1	Hyperscaler custom ASICs (TPU, Trainium, Maia, MTIA class)	PRISM-15 or PRISM-G zoned; R+ over the dies with the tray filter	samples M12–15, qualified M18–24, volume 2029
A2	NVIDIA reference designs	PRISM-G Tier 1/2; micro-channel-lid insert (Tier 3B)	through NVIDIA thermal engineering; platform 2029–30
A3	15–30 kW modules, Kyber-class blades, wafer-scale tiles	PRISM-15, -15-HF, -W	2028–29
A4	CPU, NVSwitch, co-packaged optics	PRISM-S	rack kits only
P1	800 VDC rectifier and solid-state-transformer SiC modules	PRISM-DC direct-substrate cooler	demonstrator M9–12 with a module partner; volume 2028+
P2	800 VDC shelf plates	keyed-orifice plate	rack kits only
P3	SiC traction	PRISM-A	licence 2028, vehicles 2030–31

AI data centres first. The plate is sized, zoned and tested for the 15 kW class and scales to the GPU envelope with the same core, grate and cap tooling; only the meter artwork and the pedestal change per package. The buyers who control package, tray and loop (hyperscalers) can adopt the tray filter and the R+ grade; NVIDIA is reached through its thermal engineering team, with Group-A houses as factories, not as licensees.

Power electronics second. On a 1200 V SiC half-bridge (1.62 cm² of SiC per switch, silver-sintered Si_3N_4 AMB, PG25 at 45 °C) the PRISM-DC core cuts the junction-to-coolant rise from 31.4 to 22.8 K per 400 W switch, carries 1.38× the loss per cm² of SiC at 150 °C junction, and extends power-cycling life about four to five times at equal loss ($\Delta T_j^{-4.4}$ to $^{-5}$ scaling of a 27 % smaller swing). Under a gap pad the advantage vanishes. "SSD converters" is read as SiC converters; solid-state transformers and DC breakers (15–25 kW of heat per MW on isolated substrates) fit; storage drives at 1 to 2 W/cm² do not.

13. Programme, gates, budget and decisions

13.1 Roadmap (M0 = November 2026)

Months	Work	Gate
0–1	RFQs, long-lead orders, read OCP §5.2 and the PG25 guidelines, FTO claim pulls, CBP ruling request, NDA envelope requests	–
1–3	E0 purchased plates, E1 braze coupons, E2 water table and CFD; design freeze preparation	**G1 (M3) Rev A freeze: join route, base 0.8 vs 1.0, C2 go/no-go, outline and pin reconciliation. Kill if no route passes E1 by M5**
3–5	Build A, 12 plates; E3 test-vehicle commissioning; E4 fouling starts	FTO opinion at M4: US demonstration decision

Months	Work	Gate
5–6	5 plates × 3 re-mounts, paste and pad, both vehicles	**G2 (M6): pad ≤ 31 / ≤ 36 K and paste ≤ 33 K, else re-scope the claim or pivot; at least one letter of intent**
6–9	Build B, 20 plates on production-intent dies; E5 endurance; independent laboratory	**G3 (M9): repeat within ±1.5 K**
9–12	OCP-format report; R+ and liquid-metal stretch data	**G4 (M11–12): public at OCP Global Summit 2027**
12–18	300–500-plate pilot at the contract manufacturer; PPAP; samples to two hyperscaler programmes; PRISM-G and PRISM-DC samples	**G5 (M18): qualified product**
18–36	ramp 10k to 100k per year	volume 2029

13.2 Experiments

#	Experiment	When / cost	Pass criterion
E0	Buy 2–3 incumbent GPU plates, a CoolIT 4 kW-class plate and UQD08 pairs; measure R, dP, Cv, height, mass	M0–2 / US$40–60k	rig repeatability ≤ 1 K
E1	Tie-braze matrix: kerfed coupons × routes A/B/C (+ belt C2) × base 0.8/1.0 × potted/unpotted kerfing × tip-plane tolerance × fixture pressure	M0.5–3 / US$120k	every land strip ≥ 70 % bonded over any 10 mm; unbonded strip detected at 0.8 mm; burst ≥ 4.5 MPa; flatness ≤ 20 µm
E2	Water table and single-cell CFD of orifice, lip, window, kerf	M1–3 / US$90k	segment max/min ≤ 1.15 at 18 LPM; fin-top velocity ≤ 0.5 m/s
E3	Test-vehicle commissioning (TTV-A, TTV-100, TTV-A2, TTV-25, TTV-D), D5470 interface metrology, pulsed-power transient	M2–5 / US$200k	heat capture ≥ 98 %; re-mount repeat ≤ 1 K
E4	Dose-equivalent fouling (100× ISO 12103 + CuO + EPDM, 1,000 h) with fin-field dP taps; lip erosion at 1×/1.5×/2× flow	M3–10 / US$100k	fin-field dP ≤ +10 %; dT ≤ +1 K; no measurable erosion
E5	Endurance on 22 plates; coolant chemistry; shock and vibration; orientation	M5–11 / US$250k	reliability targets of architecture 10.1
E6	Freedom-to-operate claim pulls, charts, opinion	M0–4 / US$150–250k	clear, design-around or launch-date decision

13.3 Budget and probability

Twelve months: US$7.3–8.9 M before contingency, US$8.4–10.7 M with 15 to 20 % contingency. Main lines: labour US$3.9–4.1 M (18 FTE); rigs, test vehicles and IR station US$1.4–1.6 M; confocal, helium and D5470 US$0.3–0.6 M; Build B dies US$0.25–0.6 M; prototypes US$0.35 M; reliability and chemistry US$0.38–0.5 M; IP US$0.25–0.35 M; independent laboratory US$0.15 M; CFD, FEA, CT, fit-out, compliance, E0 US$0.3–0.7 M. Tranche to G1: about US$2.3 M. Months 12–18 add US$3.0–3.8 M; the 18-month total is US$11.4–14.5 M. The lean option is about US$8.3 M with contingency and withdraws the above-18 kW and 350 W/cm² claims.

Probabilities (model × programme factors 0.8 × 0.9): decisive pad result at M12 0.43; paste average win on the parity vehicle 0.58; paste maximum win 0.65; pad average, maximum and dP together 0.68; paste average win on the 100 cm² vehicle 0.32 (R grade) or 0.50 (R+). The

dominant failure mode is a one-to-three-month slip at E1 or at test-vehicle commissioning, not a thermal miss.

13.4 Decisions requested

#	Decision	Recommendation
D1	Public claim scope	Scoped claim, released only after G2 measurement: at CoolIT's published point PRISM beats CoolIT like-for-like on average and maximum case temperature, uniformity, case-limited capacity and inlet headroom; its maximum is lower at every OCP flow; its plate pressure drop is below any credible reconstruction; lower at the bar, parity or better on mass, parity on cost per kW. It costs more per plate, ships with a tray filter and has no field record. No claim on the 4 kW point, the low-flow average or the 100 cm² average
D2	Channel and moat	Build and supply through a non-Group-A contract manufacturer to hyperscalers, and to NVIDIA through its thermal engineering team; no licence to Group-A houses
D3	Budget and gating	Tranches: US$2.3 M to G1 at M3, the rest released on a G1 pass; full scope US$8.4–10.7 M
D4	Rev A stack	Vacuum BAg braze as the baseline (US$165); the silver-free Cu-Sn belt route (US$142) decided at G1 on E1 coupons
D5	US launch and build site	No US offers, sales or evaluation units before the FTO opinion, and no US sale before 10 September 2028 unless it clears; Vietnam or Mexico build first, following a CBP binding ruling; Taiwan only after TW I606224 is charted
D6	Filtration	Tray cartridge mandatory and supplied (≥ 150 cm², 33 µm absolute); the product is "mechanically drop-in, ships with a tray filter kit"
D7	Hedges	US$0.4–0.6 M micro-channel-lid insert design study plus a two-phase watching brief

13.5 Organisation

Eighteen FTE: programme lead 1; thermal-fluids 2 (chief architect within); mechanical and FEA 2; manufacturing and supplier 2; supplier-quality engineer at the contract manufacturer 1; test laboratory 3; reliability, materials and chemistry 1.5; quality 1; business development and customer engineering 1.5; applications and OCP 1; compliance 0.5; cost engineering 0.5; IP coordination 0.5 with outside counsel. Gate board: the line manager (chair, kill authority), a finance delegate, the programme lead, the chief architect, an ex-OEM thermal architect voting on technical gates, and patent counsel advising.

14. Design status and open items at this release

Complete to design-definition level: core (DDD-core), grate (DDD-grate), plenum and separator (DDD-plenum), cap, bosses, guard and seam (DDD-cap), mounting and integration (DDD-mech), manufacturing package (DDD-mfg), parametric CAD with drawings, predicted datasheet (DDD-perf), innovation register (DDD-innovation). **Not started:** PRISM-G product artwork, PRISM-DC detail.

The cap lane reported last (`design/DDD-cap.md`): 10 × 7.7 mm arches, 35 mm cross domes with an outlet spine, bosses at y −20 and +21, a 121.5 / 123.4 skirt, 5 × 8 separator windows, an in-dome guard and a 279 g cap, at 20.8 mm to the bar. Its values are in `params.json` and the CAD is regenerated from them (section 9).

Inter-lane conflicts to close at G1 (F9, O-F4):

Item	Lane values	Recommendation
Grate and separator outline	122.6 (grate) vs 121.0 (core, CAD)	**closed by the cap lane (IR-C2): 121.0 ± 0.08, 0.25 mm inside the 121.5 skirt; the grate lane's drawing is updated at G1**
Datum pins	2.0 at ±57.5 (core, CAD) vs 3.0 at ±57 (grate)	2.0 H7 alumina; one round, one slotted
Separator thickness and height	1.8 mm (plenum lane) vs 1.5 (architecture)	keep 1.8 copper; the cap lane's 7.7 mm arches restore 20.8 mm at the bar; 0.6 mm 316L stays the mass option
Separator windows	5 × 10 (plenum) vs 5 × 8 (cap IR-C1)	5 × 8; plenum top-flat windows stay 5 × 10; re-run plenum_sep.py
Port boss positions	(0, −14) / (0, +28) provisional vs (0, −20) / (0, +21) (cap)	cap lane; inside the mechanical keep-in; mech IR-C3 to confirm the frame window
Guard	in-dome cup 17.4 cm² at 0.40 kPa (GUARD-C) vs stub-housing 25 cm² at 0.2 kPa (GUARD-A)	decide with the first customer ICD (O-C3); the plate is the same either way
Mass	1,307 (cap lane) or about 1,262 with the CAD cap mass, vs 1,240 budget and 1,400 limit; up to 1,392 with the product frame	0.6 mm 316L separator (−106 g) is the recovery; a 0.9 mm cap sheet is ruled out by the arch-end thinning limit
Ear hole	3.4 (core) vs 4.5 (mech IR-M1)	4.5 +0.10/−0
Outlet window positions	+9/+23/+37/+51 (plenum) vs +7/+21/+35/+49 (mech IR-M3)	+7/+21/+35/+49 (already in params.json and CAD); re-run plenum_sep.py
Fixture pressure	0.2 MPa (architecture) vs 0.08 (mfg)	0.08; E1 factor 0.04/0.08/0.12
Test order	architecture 5.3 vs mfg D-F4	mfg order (He dry before any wet step; IR before the fly-cut)

Open items and the test that closes each (consolidated; the lane tables hold the full list):

ID	Item	Closes at
O1	Recess depth 0.70–0.90 and lip width 0.70–0.80: fin-top velocity ≤ 0.5 m/s, jet footprint on the lip	E2 CFD and PIV, M2
O2	Orifice discharge coefficient 0.65–0.80 vs Reynolds, direction, laser vs die edge	E2 coupons, M2
O-F1	Fixture pressure vs tie-bond fraction and plenum corner set	E1, M3
O-F2	25 µm perforated vs 12–15 µm foil at the fin tips	E1 sections, M3
O-F3, O-C4	Cap draw trial on soft tools: arch-end thinning map ≥ 0.70, valley r 0.5 restrike, land flatness 0.05, skirt inner 121.50 ± 0.03	Build A, M3–4
O-C1	Outlet dome walls and top ligaments with the brazed boss ring: 28–48 MPa bound at design burst	E1 FEA, M2
O-C2	Skirt wall with a wetted stack-edge cavity; lap-seam fillet continuity by section on the first 5 plates	E1 FEA, E0
O-C3	GUARD-C in-dome cup vs GUARD-A stub housing; fouling dose on the chosen element	first customer ICD, E4
O-C6	Arch ends vs the outermost separator windows (1.3 % / 4.0 % of window area under the end wall): lengthen the arches by about 1 mm inside the 52 / 58.8 limits, or trim the windows	cap and plenum lanes, G1
O5, O6	Lamination registration ±0.03; capillary-break effectiveness; orifice area change ≤ 1 % through braze	E1 CT and vision, M3

ID	Item	Closes at
O7, O-F6	Web-to-land bypass ≤ 0.2 % of flow; dP Cpk budget	E2 dye on Build A, M4
O-F7	Flush recipe clears 10,080 kerf sites; chip CT	M4
O-M1	Test-vehicle spring rate measured; 2,011 ± 40 N	E3 rig build
O-M2	Clamp and thermal bow FEA; heated pressurised flatness scan on 5 plates; 2-side vs 4-side bearing	E1 M2, E3 M4
O-M3	Shock and vibration levels from ORv3, OCP cold-plate white paper, GR-63-CORE	M5
O-M7	Thermocouple layout confirmed against a purchased CoolIT vehicle and OCP §5.2	E0, M1
O8	Lip erosion at 2.2–2.5 m/s jets, 1,000 h	E4, M10
O9, O-F9	Perforating-die transition for the meter web after artwork freeze	M6–9
O-F10	Second-source list; CBP ruling on the build site	M6

Appendix A. File index

Path	Content
`docs/cooling/README.md`	map of the package
`docs/cooling/01-benchmark-and-requirements.md`	CoolIT reconstruction, OCP arbiter, requirements
`docs/cooling/02-physics-basis.md`	the physics the design rests on
`docs/cooling/03-architecture.md`	architecture of record, PRISM-15 Rev A (audited)
`docs/cooling/04-verification.md`	how the architecture was stress-tested
`docs/cooling/05-detailed-design.md`	design definition synthesis (this report's source for sections 3, 6, 7, 8, 14)
`docs/cooling/model/coldplate_model.py`, `model_output.md`	the calibrated 1-D / network / 2-D model, v3
`docs/cooling/design/params.json`	the CAD parameter file, single source of truth for geometry
`docs/cooling/design/DDD-core.md`, `DDD-grate.md`, `DDD-plenum.md`, `DDD-cap.md`, `DDD-mech.md`, `DDD-mfg.md`, `DDD-perf.md`, `DDD-innovation.md`	design-definition documents per lane with their scripts, parameter files and outputs
`docs/cooling/design/artwork_prism15_ttv.csv`, `.dxf`	the 15 kW orifice artwork (ART-TTV-15-A)
`docs/cooling/design/artwork_prism_g_example.csv`, `.dxf`	zoned GPU artwork example
`docs/cooling/design/bom.csv`	25-line bill of materials
`docs/cooling/design/perf_datasheet.py`, `.out`	the predicted datasheet of section 5
`docs/cooling/cad/`	parametric CAD, outputs, drawings, renders, README
`docs/cooling/report/PRISM-15-RevA-design-report.md`, `.pdf`	this report and its source

Path	Content
`docs/cooling/appendix/`	research, physics, design-panel, verification and judge evidence with scripts

Appendix B. Conventions

Units are millimetres unless stated. Temperatures in °C, differences in K. PG25 is 25 % propylene glycol in water by volume, inhibited, with conductivity 0.49 W/m·K at 45 °C. "Case-to-inlet" is the heater-surface or package-case temperature minus the plate inlet coolant temperature, as the OCP test specification defines it, with the thermal interface included. R is that difference divided by power (K/W); R'' is R multiplied by the heated area (cm²·K/W). "CoolIT measured" values are the three figures in CoolIT's June 2026 tech brief; "CoolIT-like" values are our calibrated model of a split-flow plate and are not CoolIT's numbers. Axes: X along the fins (across the fifteen lines), Y along the lines, Z up from the die face, origin at the centre of the fin field.